

4. COMPLEX NUMBERS

SYNOPSIS

INFORMATION :

- * The quantity $\sqrt{-1}$ denoted by i , is called Imaginary unit.

$$i^2 = \sqrt{-1} \times \sqrt{-1} \neq 1, \text{ but equals to } -1.$$

$$i^n = \begin{cases} (-1)^{n/2} & \text{if } n \text{ is even integer} \\ (-1)^{\frac{n-1}{2}} & \text{if } n \text{ is odd integer} \end{cases}$$

$$\sqrt{a}\sqrt{b} = \sqrt{ab} \text{ holds only when atleast one of } a, b \text{ is non-negative}$$

- * If $a, b \in \mathbb{R}$, $a+ib$ is called complex number usually denoted by Z

$$Z \text{ is } \begin{cases} \text{purely real if } \operatorname{Im}(Z) = 0 \\ \text{purely imaginary if } \operatorname{Re}(Z) = 0 \\ \text{imaginary if } \operatorname{Im}(Z) \neq 0 \end{cases}$$

If Z_1, Z_2 are two complex numbers (which are not purely real) then either $Z_1 = Z_2$ or $Z_1 \neq Z_2$ only holds.

$Z_1 > Z_2$, $Z_1 < Z_2$ are meaningless i.e. **there is no order among complex numbers.**

- * Every Complex number $Z = a+ib$ can be represented by an ordered pair (a,b) and hence can be represented by a point with co-ordinates (a,b) in the complex plane (Argand diagram).

* Modulus & Argument of 'Z'

Let $Z = x+iy$, then

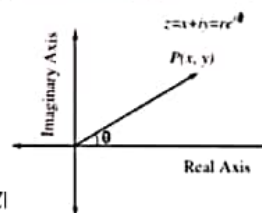
$$|Z| = \text{Distance of } Z \text{ from origin} = \sqrt{x^2 + y^2}$$

$$|Z| \geq 0 \text{ but } |Z| = \begin{cases} z & \text{for } z > 0 \\ -z & \text{for } z < 0 \end{cases} \text{ is not correct.}$$

Let $Z = x+iy$ be represented by P , then the length of $\overline{OP}$ is $|Z|$

If OP makes angle ' θ ' with positive direction of real axis, then ' θ ' is argument or amplitude of Z

$$\theta = \tan^{-1}\left(\frac{y}{x}\right)$$



* Conjugate of Z is denoted by $\bar{Z}$.

$$\text{If } Z = a+ib, \bar{Z} = a-ib$$

$$\text{Note that } Z + \bar{Z} = 2 \operatorname{Re}(Z)$$

$$Z - \bar{Z} = 2i \operatorname{Im}(Z)$$

$$Z \bar{Z} = (\operatorname{Re}(Z))^2 + (\operatorname{Im}(Z))^2 = |Z|^2$$

$$\text{if } Z_1 = x_1+iy_1, Z_2 = x_2+iy_2, \text{ then } \operatorname{Re}(Z_1 \bar{Z}_2) = \operatorname{Re}(\bar{Z}_1 Z_2) = x_1 x_2 + y_1 y_2$$

$$Z \text{ is purely real} \Leftrightarrow Z = \bar{Z}$$

$$Z \text{ is purely Imaginary} \Leftrightarrow Z = -\bar{Z}$$

Geometrically, ' $\bar{Z}$ ' is reflection of ' Z ' in real axis. $Z, \bar{Z}, -Z, -\bar{Z}$ in order form rectangle.

* **Properties of Conjugate of Z**

$$Z_1 + Z_2 + \dots + Z_n = \overline{Z_1} + \overline{Z_2} + \dots + \overline{Z_n}$$

$$Z_1 Z_2 \dots Z_n = \overline{Z_1} \overline{Z_2} \dots \overline{Z_n}$$

$$\left(\frac{Z_1}{Z_2} \right) = \frac{\overline{Z_1}}{\overline{Z_2}}$$

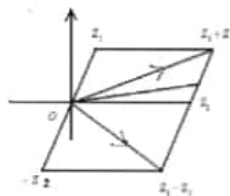
If $f(x+iy) = a+ib$, then $f(x-iy) = a-ib$

$f(z) = g(z)$ then $f(\overline{z}) = g(\overline{z})$

* **Geometrical meaning of addition & subtraction of complex numbers**

Given the points Z_1 and Z_2 , complete the parallelogram with OZ_1 and OZ_2 as adjacent sides, where O is the origin. Then the 4th vertex of this parallelogram is $Z_1 + Z_2$

Like wise $Z_1 - Z_2$ is the 4th vertex of parallelogram having $O, Z_1 - Z_2$ as three of its vertices



We can see that $Z_1 + Z_2$ & $Z_1 - Z_2$ represent diagonals of the parallelogram having Z_1, Z_2 as adjacent sides like $\vec{a} + \vec{b}, \vec{a} - \vec{b}$ represent diagonals of the parallelogram having $\vec{a}, \vec{b}$ as adjacent sides

* **Complex number as a vector**

If $z = x+iy$ is represented by 'P' then $\overrightarrow{OP}$ is position vector of P w.r.t. 'O' represents 'Z'. Magnitude of vector $\overrightarrow{OP}$ is given by $|Z|$, direction of vector $\overrightarrow{OP}$ is given by $\arg Z$.

If P, Q represent the complex numbers Z_1, Z_2 then the vector $\overrightarrow{PQ}$ represents $Z_2 - Z_1$, since $\overrightarrow{PQ} = \overrightarrow{OQ} - \overrightarrow{OP} = Z_2 - Z_1$.

* **Properties of Moduli**

- * $-|Z| \leq \operatorname{Re}(Z) \leq |Z|$
- * $-|Z| \leq \operatorname{Im}(Z) \leq |Z|$
- * $|Z| \leq |\operatorname{Re}(Z)| + |\operatorname{Im}(Z)| \leq \sqrt{2}|Z|$
- * $|Z| = |\operatorname{Re}(Z)| + |\operatorname{Im}(Z)|$ if Z is purely real or purely imaginary
- * $\sqrt{2}|Z| = \operatorname{Re}(Z) + \operatorname{Im}(Z)$ if $|\operatorname{Re}(Z)| = |\operatorname{Im}(Z)|$
- * $|Z_1 Z_2 Z_3 \dots Z_n| = |Z_1| |Z_2| \dots |Z_n|$
- * $\left| \frac{Z_1}{Z_2} \right| = \frac{|Z_1|}{|Z_2|}$ if $Z_2 \neq 0$

- * $\|Z_1\| - \|Z_2\| \leq \|Z_1 + Z_2\| \leq \|Z_1\| + \|Z_2\|$ (Triangle inequality)

$\|Z_1 + Z_2\| = \|Z_1\| + \|Z_2\|$ holds if 0, Z_1 , Z_2 are collinear and Z_1 , Z_2 lie on same side of origin

i.e when $\arg Z_1 - \arg Z_2 = 0$ i.e $\frac{Z_1}{Z_2} > 0$

$\|Z_1 - Z_2\| = \|Z_1\| + \|Z_2\|$ holds if 0, Z_1 , Z_2 are collinear and Z_1 , Z_2 lie on opposite side of origin

i.e when $\arg Z_1 - \arg Z_2 = \pi$ i.e $\frac{Z_1}{Z_2} < 0$

$\|Z_1 - Z_2\| = \left| \|Z_1\| - \|Z_2\| \right|$ holds if 0, Z_1 , Z_2 are collinear and Z_1 , Z_2 lie on same side of origin

$\|Z_1 + Z_2\| = \left| \|Z_1\| - \|Z_2\| \right|$ holds if 0, Z_1 , Z_2 are collinear and Z_1 , Z_2 lie on opposite side of origin

- * In general, $\|Z_1 + Z_2 + \dots + Z_n\| \leq \|Z_1\| + \|Z_2\| + \dots + \|Z_n\|$

Equality holds when 0, Z_1 , $Z_2, \dots, Z_n$ are collinear and $Z_1, Z_2, Z_3, \dots, Z_n$ lie on same side of origin

- * $|Z_1 + Z_2|^2 = (Z_1 + Z_2)(\overline{Z_1} + \overline{Z_2}) = |Z_1|^2 + |Z_2|^2 + 2\operatorname{Re}(Z_1 \overline{Z_2})$

- * $|Z_1 - Z_2|^2 = (Z_1 - Z_2)(\overline{Z_1} - \overline{Z_2}) = |Z_1|^2 + |Z_2|^2 - 2\operatorname{Re}(Z_1 \overline{Z_2})$

- * $|Z_1 + Z_2|^2 + |Z_1 - Z_2|^2 = 2(|Z_1|^2 + |Z_2|^2)$ (Parallelogram law)

- * If $a, b \in \mathbb{R}$ and Z_1, Z_2 are complex numbers then

$$|aZ_1 + bZ_2|^2 + |bZ_1 - aZ_2|^2 = (a^2 + b^2)(|Z_1|^2 + |Z_2|^2)$$

- * If θ is an argument of Z , then $2n\pi + \theta, n \in \mathbb{Z}$ is general argument of Z

The principal argument of Z lies in the interval $(-\pi, \pi]$

Let $Z = x + iy$

Step - I : Find $\theta = \tan^{-1}\left(\frac{y}{x}\right)$ then $\theta \in \left[0, \frac{\pi}{2}\right)$

Step - II : Find the quadrant in which Z lies from the signs of x & y coordinates

Then $\arg(Z)$ will be θ or $\pi - \theta$ or $\theta - \pi$ or $-\theta$ according as Z lies in I or II or III or IV quadrants

If $Z = 0$, $\arg Z$ is not defined

- * **Properties of $\arg(Z)$**

$$\operatorname{Arg} Z = \theta \Rightarrow \operatorname{Arg} \overline{Z} = -\theta, \operatorname{Arg}\left(\frac{1}{Z}\right) = -\theta, \operatorname{Arg}\left(\frac{1}{\overline{Z}}\right) = \theta \text{ if } Z \text{ is not purely real.}$$

$$\operatorname{Arg}(Z_1 Z_2) = \operatorname{Arg} Z_1 + \operatorname{Arg} Z_2 + 2k\pi \quad k \in \{0, 1, -1\}$$

$$\operatorname{Arg}\left(\frac{Z_1}{Z_2}\right) = \operatorname{Arg} Z_1 - \operatorname{Arg} Z_2 + 2k\pi \quad k \in \mathbb{I}$$

$$\operatorname{Arg}(Z^n) = n \arg(Z) + 2k\pi \quad k \in \{0, 1, -1\}$$

Proper value of K must be chosen so that RHS also lies in $(-\pi, \pi]$

For eg:-

If $Z_1 = \text{cis } \frac{2\pi}{5}$, $Z_2 = \text{cis } \frac{4\pi}{5}$ then $Z_1 Z_2 = \text{cis } \left(\frac{6\pi}{5} \right)$

$\text{Arg}(Z_1 Z_2) = \frac{6\pi}{5}$ since $Z_1 Z_2 \in Q_3$ but $\text{Arg} Z_1 + \text{Arg} Z_2 = \frac{6\pi}{5}$ so if K is chosen as '-1'

then $\text{RHS} = \frac{6\pi}{5} - 2\pi$ hence both sides will be same

If $|Z| = r$, $\text{Arg}(Z) = \theta$ then $Z = r(\cos \theta + i \sin \theta)$ is called trigonometric form or polar form of complex number Z. Here we should take Principal value of $\arg(Z)$

If $|Z| = r$, $\text{Arg}(Z) = \theta$, then $Z = re^{i\theta}$ is called exponential form or Euler's form of Z

* Rotation Theorem

If Z_1, Z_2 are two complex numbers then $\arg \left(\frac{Z_1}{Z_2} \right)$ is the angle through which OZ_2 must be turned in order that it coincides with OZ_1

$$\frac{Z_1}{Z_2} = \frac{|Z_1|e^{i\theta_1}}{|Z_2|e^{i\theta_2}} = \frac{|Z_1|}{|Z_2|}e^{i(\theta_1 - \theta_2)}$$



In general, if Z_1, Z_2, Z_3 be three vertices of $\triangle ABC$ described in the Counter-Clock wise sense and $\angle CAB = \alpha$

* **Demoiver's Theorem:** If $n \in \mathbb{N}$, $(\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta$

* If $n \in \mathbb{Q}$ (set of rationals) then $\cos n\theta + i \sin n\theta$ is one of the values of $(\cos \theta + i \sin \theta)^n$. This theorem is useful in finding the roots of any complex number

* nth root of unity are Solutions of $Z^n = 1$. They are given by $\text{Cis} \left(\frac{2r\pi}{n} \right)$, $r = 0, 1, 2, \dots, n-1$

* If $\text{Cis} \left(\frac{2\pi}{n} \right)$ is defined by α then $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$ are n^{th} roots of unity. The points representing these n^{th} roots are located at the vertices of a regular polygon of 'n' sides inscribed in a unit circle having centre at the origin. One vertex being on +ve real axis

* $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$ are in G.P with Common ratio $e^{i\frac{2\pi}{n}}$

$$1 + \alpha + \alpha^2 + \dots + \alpha^{n-1} = \left(\frac{1 - \alpha^n}{1 - \alpha} \right) = 0$$

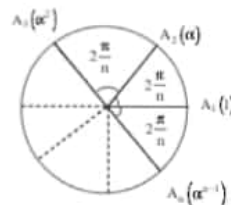
$$\Rightarrow \sum_{r=0}^{n-1} \cos \left(\frac{2r\pi}{n} \right) = 0, \sum_{r=0}^{n-1} \sin \left(\frac{2r\pi}{n} \right) = 0$$

$$1, \alpha, \alpha^2, \dots, \alpha^{n-1} = \alpha^{\frac{n(n-1)}{2}} = \left(\text{cis } \frac{2\pi}{n} \right)^{\frac{n(n-1)}{2}} = (\text{cis } \pi)^{n-1}$$

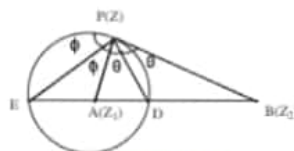
Thus product of nth root of unity $= (-1)^{n-1} \begin{cases} -1, & \text{if } n \text{ is even} \\ 1, & \text{if } n \text{ is odd} \end{cases}$

* If n^{th} roots of unity are denoted by $1, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$ then

$$1^m + \alpha_1^m + \alpha_2^m + \dots + \alpha_{n-1}^m = \begin{cases} 0 & \text{if } m \text{ is not integral multiple of } n \\ n & \text{if } m \text{ is integral multiple of } n \end{cases} \Rightarrow |\alpha_i - \alpha_j| = 2 \sin \left| \frac{(i-j)\pi}{n} \right|$$



- * If $1, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are roots of $Z^n - 1 = 0$ then $\alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are roots of $Z^{n-1} + Z^{n-2} + \dots + Z + 1 = 0$. Hence we have
 $(Z - \alpha_1)(Z - \alpha_2) \dots (Z - \alpha_{n-1}) = Z^{n-1} + Z^{n-2} + \dots + Z + 1$ and hence $(1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1}) = n$
 $(1 + \alpha_1)(1 + \alpha_2) \dots (1 + \alpha_{n-1}) = \begin{cases} 0 & \text{if } n \text{ is even} \\ 1 & \text{if } n \text{ is odd} \end{cases}$
- * $Z^n = P^n$ has roots $P, P\alpha_1, P\alpha_2, \dots, P\alpha_{n-1}$ where α_i 's are n^{th} roots of unity
- * Product of all values of $(z_1)^{1/n}$ is $(-1)^{n-1} z_1$
- * **Cube roots of Unity :** Twelfth roots of unity are
 $1, \frac{-1+i\sqrt{3}}{2}, \frac{-1-i\sqrt{3}}{2}$ are roots of $Z^3 = 1$ $\pm 1, \pm W, \pm W^2$
 If $\frac{-1+i\sqrt{3}}{2} = w$, then $\frac{-1-i\sqrt{3}}{2} = w^2$ or $\bar{w}$ or $\frac{1}{w}$ $\pm i, \pm iW, \pm iW^2$
 Also $w^3 = 1, 1 + w + w^2 = 0$
- * In general, $1 + w^m + w^{2m} = \begin{cases} 0 & \text{if } m \in I, \text{ but not integral multiple of } 3 \\ 3 & \text{if } m \text{ is integral multiple of } 3 \end{cases}$
- * $1, w, w^2$ are the vertices of an equilateral triangle inscribed in the circle $|Z|=1$
- * If a, b, c are non zero numbers such that $a+b+c = a^2+b^2+c^2$ then $a:b:c = 1:w:w^2$
 The linear factors of
 i) x^3+y^3 are $(x+y), (x+wy), (x+w^2y)$ ii) x^3-y^3 are $(x-y), (x-wy), (x-w^2y)$
 iii) $x^2+y^2+z^2-xy-yx-zx$ are $(x+wy+w^2z), (x+w^2y+wz)$
 iv) $x^3+y^3+z^3-3xyz$ are $(x+y+z), (x+wy+w^2z), (x+w^2y+wz)$
- * **Different Loci :**
- * $|Z_1 - Z_2|$ represents distance between the points Z_1, Z_2
- * Locus of Z satisfying $|Z - Z_0| = r$ is circle with centre at Z_0 and of radius r
 $|Z - Z_0| < r, |Z - Z_0| > r$ represents the regions inside and outside the circle $|Z - Z_0| = r$ respectively
- * $r_1 < |Z - Z_0| < r_2$ represent the region of argand plane lying between the two concentric circles $|Z - Z_0| = r_1, |Z - Z_0| = r_2$
- * Locus of Z satisfying $|Z - Z_1| = |Z - Z_2|$ is perpendicular bisector of line segment joining Z_1, Z_2
 If Z is such that $|Z - Z_1| = |Z - Z_2| = |Z - Z_3|$ then Z is circumcentre of triangle formed by the vertices Z_1, Z_2, Z_3
- * If $\left| \frac{Z - Z_1}{Z - Z_2} \right| = K (K \neq 1, > 0)$, the locus of Z is Apollonius circle with $\frac{KZ_2 + Z_1}{K + 1}, \frac{KZ_2 - Z_1}{K - 1}$ as diametrical ends and with radius $\frac{K|Z_1 - Z_2|}{|K^2 - 1|}$

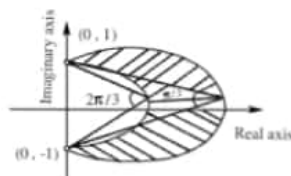


- * Locus of Z satisfying $\operatorname{Re}\left(\frac{Z-Z_1}{Z-Z_2}\right)=0$ or $\frac{Z-Z_1}{Z-Z_2}=\lambda i$ or $\operatorname{Arg}\left(\frac{Z-Z_1}{Z-Z_2}\right)=\pm\frac{\pi}{2}$ is a circle with Z_1, Z_2 as diametrical end points

$\operatorname{Arg}\left(\frac{Z-Z_1}{Z-Z_2}\right)=\frac{\pi}{2} \Rightarrow$ Locus of ' Z ' is semi-circle with Z_1, Z_2 as extremities (Locus excludes the points Z_1, Z_2)

If $\left|\operatorname{Arg}\left(\frac{Z-Z_1}{Z-Z_2}\right)\right|=\frac{\pi}{2}$, Locus of Z is a circle with Z_1, Z_2 as ends of diameter.

- * $\operatorname{Arg}\left(\frac{Z-Z_1}{Z-Z_2}\right)=\theta \Rightarrow$ Locus of Z is major arc or minor arc of circle with Z_1, Z_2 as end points of a chord of circle (Locus excludes the points Z_1, Z_2) according as $\theta \in \left(0, \frac{\pi}{2}\right)$ or $\theta \in \left(\frac{\pi}{2}, \pi\right)$
- * Suppose Z is such that $\frac{\pi}{3} \leq \arg\left(\frac{Z+i}{Z-i}\right) \leq \frac{2\pi}{3}$. Then locus of Z is shaded region in adjacent fig (excluding the points $i, -i$)



- * Locus of Z satisfying $\operatorname{Im}\left(\frac{Z-Z_1}{Z-Z_2}\right)=0, \operatorname{Arg}\left(\frac{Z-Z_1}{Z-Z_2}\right)=0$ or π is the line passing through Z_1, Z_2

$\operatorname{Arg}\left(\frac{Z-Z_1}{Z-Z_2}\right)=\begin{cases} 0 & \Rightarrow \text{Locus of } Z \text{ is the line joining } Z_1, Z_2 \text{ excluding the line segment } \overline{Z_1 Z_2} \\ \pi & \Rightarrow \text{Locus of } Z \text{ is the segment joining } Z_1, Z_2 \end{cases}$

- * Locus of Z satisfying $\operatorname{Arg}(Z-Z_1)=\alpha$ is a ray with vertex at Z_1 and making angle ' α ' with +ve direction of real axis (Locus excludes the point Z_1)
- * Locus of Z satisfying $|Z-Z_1|+|Z-Z_2|=2a$ when $2a > |Z_1-Z_2|$ is an ellipse with Z_1, Z_2 as foci and ' $2a$ ' as length of major axis and eccentricity $e=\frac{|Z_1-Z_2|}{2a}$. When $2a = |Z_1-Z_2|$ Locus of Z is the line segment joining Z_1, Z_2 . There can not be any point in the argand plane such that $2a < |Z_1-Z_2|$
- * Locus of Z satisfying $||Z-Z_1|-|Z-Z_2||=2a$ is a hyperbola if $2a < |Z_1-Z_2|$, its foci are Z_1, Z_2 & $e=\frac{|Z_1-Z_2|}{2a}$. In case, $|Z_1-Z_2|=2a$, locus of Z is the line $\overline{Z_1 Z_2}$ excluding the line segment joining Z_1 and Z_2
- * Locus of Z satisfying $|Z-Z_1|^2+|Z-Z_2|^2=K$ is a circle if $k \geq \frac{1}{2}|Z_1-Z_2|^2$

* Geometrical Applications :

- * If l, m, n are three real numbers such that $lZ_1 + mZ_2 + nZ_3 = 0$ and $l + m + n = 0$, l, m, n are not all zeros simultaneously, then the points Z_1, Z_2, Z_3 are collinear

- * The points Z_1, Z_2, Z_3 are collinear if $\frac{Z_3 - Z_1}{Z_2 - Z_1} = \frac{\bar{Z}_3 - \bar{Z}_1}{\bar{Z}_2 - \bar{Z}_1}$ or $\sum \bar{Z}_i (Z_j - Z_k) = 0$

$$\text{or } \begin{vmatrix} Z_1 & \bar{Z}_1 & 1 \\ Z_2 & \bar{Z}_2 & 1 \\ Z_3 & \bar{Z}_3 & 1 \end{vmatrix} = 0$$

- * The equation of line joining Z_1, Z_2 in different form

- 1) $Z = Z_1 + t(Z_2 - Z_1)$ 't' being parameter (parametric form)

$$2) \begin{vmatrix} Z & \bar{Z} & 1 \\ Z_1 & \bar{Z}_1 & 1 \\ Z_2 & \bar{Z}_2 & 1 \end{vmatrix} = 0 \text{ (determinant form)}$$

$$3) Z(\bar{Z}_1 - \bar{Z}_2) - \bar{Z}(Z_1 - Z_2) = \bar{Z}_1 Z_2 - Z_1 \bar{Z}_2$$

This equation can put in the form $\bar{\alpha}Z + \alpha\bar{Z} + \beta = 0$ where β is real and α is a non zero complex constant $\bar{\alpha}Z + \alpha\bar{Z} + \beta = 0$ is the general equation of line in complex plane

Equation of line parallel to the line $\bar{\alpha}Z + \alpha\bar{Z} + \beta = 0$ is $\bar{\alpha}Z + \alpha\bar{Z} + \gamma = 0$, γ is a real number

Equation of line perpendicular to the line $\bar{\alpha}Z + \alpha\bar{Z} + \beta = 0$ is $\bar{\alpha}Z - \alpha\bar{Z} + i\gamma = 0$ where γ is a real number

- * Perpendicular bisector of the line segment joining Z_1 and Z_2 is

$$(\bar{Z}_1 - \bar{Z}_2)Z + (Z_1 - Z_2)\bar{Z} = |Z_1|^2 - |Z_2|^2$$

- * The equation of circle with centre at Z_0 and with radius r is

$$|Z - Z_0| = r \Rightarrow (Z - Z_0)(\bar{Z} - \bar{Z}_0) = r^2 \Rightarrow Z\bar{Z} - Z_0\bar{Z} - \bar{Z}_0Z + Z_0\bar{Z}_0 - r^2 = 0$$

which is of the form $Z\bar{Z} + \bar{\alpha}Z + \alpha\bar{Z} + \beta = 0$ (called general equation of circle)

The centre of this circle is ' $-\alpha$ ', radius is $\sqrt{\alpha\bar{\alpha} - \beta}$ ($\beta \in \mathbb{R}$)

- * The equation of circle described on the line segment joining Z_1 and Z_2 as diameter is

$$(Z - Z_1)(\bar{Z} - \bar{Z}_2) + (Z - Z_2)(\bar{Z} - \bar{Z}_1) = 0$$

- * The condition for four points Z_1, Z_2, Z_3 & Z_4 to be concyclic is, the number $\frac{Z_3 - Z_1}{Z_3 - Z_2} \cdot \frac{Z_4 - Z_2}{Z_4 - Z_1}$ is real

Hence the equation of a circle through 3 non collinear points Z_1, Z_2 & Z_3 can be taken as

$$\left(\frac{Z - Z_3}{Z - Z_1} \right) \left(\frac{Z_3 - Z_1}{Z_3 - Z_2} \right) \text{ is real } \Rightarrow \frac{(Z - Z_3)(Z_3 - Z_1)}{(Z - Z_1)(Z_3 - Z_2)} = \frac{(\bar{Z} - \bar{Z}_3)(\bar{Z}_3 - \bar{Z}_1)}{(\bar{Z} - \bar{Z}_1)(\bar{Z}_3 - \bar{Z}_2)}$$

- * Let Z_1, Z_2, Z_3 be vertices of a ΔABC , then area of $\Delta ABC = \frac{1}{4} \left| \sum \overline{Z_1}(Z_2 - Z_3) \right|$

$$\text{Centroid} = \frac{Z_1 + Z_2 + Z_3}{3}; \quad \text{Circumcentre} = \frac{\sum |Z_i|^2 (Z_2 - Z_3)}{\sum \overline{Z_1}(Z_2 - Z_3)}$$

$$\text{Orthocentre} = \frac{\sum \overline{Z_1}(Z_2 - Z_3)(Z_2 + Z_3 - Z_1)}{\sum \overline{Z_1}(Z_2 - Z_3)}$$

- * If Z_1, Z_2, Z_3, Z_4 are vertices of a quadrilateral then

$$|Z_1 - Z_3||Z_2 - Z_4| \leq |Z_1 - Z_2||Z_3 - Z_4| + |Z_1 - Z_4||Z_2 - Z_3|$$

Equality holds if and only if it is cyclic quadrilateral

- * If $Z_1, Z_2, \dots, Z_n$ are vertices of n sided regular polygon and Z_0 is its centroid then

$$\text{i) } \sum_{i=1}^n Z_i = nZ_0 \quad \text{ii) } \sum_{i=1}^n Z_i^2 = nZ_0^2$$

- * If Z_1, Z_2, Z_3 are vertices of equilateral triangle then

$$\text{i) } (Z_1 - Z_2)^2 + (Z_2 - Z_3)^2 + (Z_3 - Z_1)^2 = 0 \quad \text{ii) } \sum \frac{1}{Z_1 - Z_2} = 0$$

$$\text{iii) } \sum Z_i^2 = \sum Z_i Z_j \quad \text{iv) } (Z_1 - Z_2)^2 = (Z_2 - Z_3)(Z_3 - Z_1)$$

Logarithm of a complex number : $\log_e Z = \log_e |Z| + i(2n\pi + \arg Z)$ where $n \in I$

DeMoivre's theorem, n^{th} roots of unity etc.,

1. i) $(\text{cis } \alpha)(\text{cis } \beta) = \text{cis } (\alpha + \beta)$ ii) $(\text{cis } \alpha)/(\text{cis } \beta) = \text{cis } (\alpha - \beta)$

2. **DeMoivre's theorem for integral index :**

If n is an integer, then

$$\text{i) } (\cos \theta + i \sin \theta)^n = \cos n\theta + i \sin n\theta \quad \text{ii) } (\cos \theta + i \sin \theta)^{-n} = \cos n\theta - i \sin n\theta$$

$$\text{iii) } (\cos \theta - i \sin \theta)^n = \cos n\theta - i \sin n\theta \quad \text{iv) } (\cos \theta - i \sin \theta)^{-n} = \cos n\theta + i \sin n\theta$$

$$\text{v) } \cos n\theta + i \sin n\theta = \frac{1}{\cos n\theta - i \sin n\theta} \Rightarrow \cos n\theta - i \sin n\theta = \frac{1}{\cos n\theta + i \sin n\theta}$$

$$\text{vi) } (\sin \theta + i \cos \theta)^n = i^n (\cos \theta - i \sin \theta)^n = i^n (\cos n\theta - i \sin n\theta)$$

$$\text{vii) } \sin \theta - i \cos \theta = -i \text{cis } \theta \quad \text{viii) } (\sin \theta - i \cos \theta)^n = (-i)^n \text{cis } n\theta$$

$$\text{ix) } 1 + \text{cis } \theta = 2 \cos \frac{\theta}{2} \text{cis } \frac{\theta}{2} \quad \text{x) } 1 + \text{cis } (-\theta) = 2 \cos \frac{\theta}{2} \text{cis } \left(\frac{-\theta}{2} \right)$$

$$\text{xi) } \frac{1 + \text{cis } \theta}{1 + \text{cis } (-\theta)} = \text{cis } \theta \quad \text{xii) } \text{cis } \theta - 1 = 2i \sin \frac{\theta}{2} \text{cis } \frac{\theta}{2}$$

$$\text{xiii) } \frac{\text{cis } \theta - 1}{\text{cis } \theta + 1} = i \tan \frac{\theta}{2}$$

3. **n^{th} root of a complex number :** The n^{th} roots of a complex number $z = r \text{cis } \theta$ are $r^{1/n} \text{cis } \left(\frac{2k\pi + \theta}{n} \right)$ where $k = 0, 1, 2, \dots, n-1$

4. ***n*th roots of unity** : The *n*th roots of unity are $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$ where $\alpha = \text{cis} \frac{2\pi}{n}$
- The *n*th roots of unity form a G.P with common ratio $\text{cis} \frac{2\pi}{n}$
 - The sum of *n*th roots of unity = 0 if $n > 2$
 - The product of *n*th roots of unity = $(-1)^{n-1}$
 - Sum of *n*th roots of any complex number *z* is 0 if $n > 2$
 - Product of *n*th roots of any complex number *z* is $z(-1)^{n-1}$
 - n*th roots of any unity are the vertices of a regular polygon of *n* sides which is inscribed in the unit circle with centre at origin and length of each side $2 \sin \frac{\pi}{n}$ and Area $n \sin \frac{\pi}{n} \cos \frac{\pi}{n}$
 - If $1, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are *n*th roots of unity then $|\alpha_i - \alpha_j| = 2 \sin \left| \frac{(i-j)\pi}{n} \right|$
 - Cube roots of unity are $1, \text{cis} \frac{2\pi}{3}, \text{cis} \frac{4\pi}{3}$, there are generally denoted by $1, w, w^2$.
 $1, w, w^2$ are the vertices of an equilateral triangle which is inscribed in a unit circle with centre at origin length of its side is $\sqrt{3}$ and area is $\frac{3\sqrt{3}}{4}$ sq.units
 - Fourth roots of unity are $1, i, -1, -i$. These are denoted by $\text{cis} 0, \text{cis} \frac{\pi}{2}, \text{cis} \pi$ and $\text{cis} \left(\frac{3\pi}{2} \right)$ these are the vertices of a square inscribed in a unit circle with centre at origin and length of each side is $\sqrt{2}$ units and Area 2 square units
5. If $\cos \alpha + \cos \beta + \cos \gamma = 0 = \sin \alpha + \sin \beta + \sin \gamma$ then
- $\cos 3\alpha + \cos 3\beta + \cos 3\gamma = 3 \cos(\alpha + \beta + \gamma)$
 - $\sin 3\alpha + \sin 3\beta + \sin 3\gamma = 3 \sin(\alpha + \beta + \gamma)$
 - $\cos 2\alpha + \cos 2\beta + \cos 2\gamma = 0$
 - $\sin 2\alpha + \sin 2\beta + \sin 2\gamma = 0$
 - $\cos 2^n \alpha + \cos 2^n \beta + \cos 2^n \gamma = 0$
 - $\sin 2^n \alpha + \sin 2^n \beta + \sin 2^n \gamma = 0$
 - $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 3/2$
 - $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 3/2$
 - $\cos(2\alpha - \beta - \gamma) + \cos(2\beta - \gamma - \alpha) + \cos(2\gamma - \alpha - \beta) = 3$
 - $\sin(2\alpha - \beta - \gamma) + \sin(2\beta - \gamma - \alpha) + \sin(2\gamma - \alpha - \beta) = 0$

6. ***Demoivre's Theorem for rational index:***

If $\frac{p}{q}$ ($q > 1$) is a rational number then $\cos \frac{p\theta}{q} + i \sin \frac{p\theta}{q}$ is one root of the q^{th} roots of $(\cos \theta + i \sin \theta)^p$

Amp. mod-Amp forms & Euler forms :

$$\begin{aligned}
 1 &= \text{cis} 0 = e^{i0} & i &= \text{cis} \frac{\pi}{2} = e^{i\frac{\pi}{2}} & -1 &= \text{cis} \pi = e^{i\pi} & -i &= \text{cis} \left(-\frac{\pi}{2} \right) = e^{-i\frac{\pi}{2}} \\
 1+i &= \sqrt{2} \text{cis} \frac{\pi}{4} = \sqrt{2} e^{i\frac{\pi}{4}} & 1-i &= \sqrt{2} \text{cis} \left(-\frac{\pi}{4} \right) = \sqrt{2} e^{-i\frac{\pi}{4}} \\
 -1+i &= \sqrt{2} \text{cis} \left(\frac{3\pi}{4} \right) = \sqrt{2} e^{i\frac{3\pi}{4}} & -1-i &= \sqrt{2} \text{cis} \left(-\frac{3\pi}{4} \right) = \sqrt{2} e^{-i\frac{3\pi}{4}}
 \end{aligned}$$

LECTURE SHEET

EXERCISE-I

*Modulus, Argument, Conjugate of Complex Numbers SQRT, Cube Root;
Problems on modulus; Problems on Argument*

LEVEL-I (MAIN)

Single answer type questions

1. If $\frac{3}{2 + \cos\theta + i\sin\theta} = x + iy$ then $(x-1)(y-3) =$
 - 1) y^2
 - 2) $-y^2$
 - 3) 0
 - 4) 1
2. If $(x + iy)^{\frac{1}{3}} = a + ib$ then $\left(\frac{x}{a}\right) + \left(\frac{y}{b}\right) =$
 - 1) $a^2 - b^2$
 - 2) $4(a^2 - b^2)$
 - 3) $2(a^2 - b^2)$
 - 4) $2(a^2 + b^2)$
3. If $\frac{\sin(x/2) + \cos(x/2) + i \tan x}{1 + 2i \sin(x/2)}$ is real then $x =$
 - 1) $n\pi$ or $n\pi + \frac{\pi}{4}$
 - 2) $2n\pi$ or $n\pi + \frac{\pi}{4}$
 - 3) $n\pi$ or $n\pi + \frac{\pi}{2}$
 - 4) $n\pi$ or $n\pi + \frac{\pi}{6}$
4. If $(1-i)(1-2i)\dots\dots(1-ni) = x - iy$, then 2, 5, 10, ... $(1+n^2) =$
 - 1) $x^2 + y^2$
 - 2) $x + y$
 - 3) $x^2 - y^2$
 - 4) $x - y$
5. If $(1 + \cos\theta - i \sin\theta)(1 + \cos 2\theta + i \sin 2\theta) = x + iy$ then $x^2 + y^2 =$
 - 1) $16 \cos^2 \theta \sin^2 \left(\frac{\theta}{2}\right)$
 - 2) $16 \sin^2 \theta \cos^2 \left(\frac{\theta}{2}\right)$
 - 3) $16 \sin^2 \theta \sin^2 \left(\frac{\theta}{2}\right)$
 - 4) $16 \cos^2 \theta \cos^2 \left(\frac{\theta}{2}\right)$
6. If $|z_1| = |z_2| = |z_3| = \dots\dots = |z_n| = 1$, then $|z_1 + z_2 + z_3 + \dots\dots + z_n| =$
 - 1) $\left| \frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} + \dots + \frac{1}{z_n} \right|$
 - 2) $\left| \frac{1}{z_1} - \frac{1}{z_2} + \frac{1}{z_3} - \dots + \frac{1}{z_n} \right|$
 - 3) $\left| \frac{1}{z_1^2} + \frac{1}{z_2^2} + \frac{1}{z_3^2} + \dots + \frac{1}{z_n^2} \right|$
 - 4) $\left| \frac{1}{z_1^2} - \frac{1}{z_2^2} + \frac{1}{z_3^2} - \dots + \frac{1}{z_n^2} \right|$
7. If z_1, z_2 are two complex numbers satisfying $\left| \frac{z_1 - 3z_2}{3 - z_1 z_2} \right| = 1$, $|z_1| \neq 3$, then $|z_2| =$
 - 1) 1
 - 2) 2
 - 3) 3
 - 4) 4

LECTURE SHEET

❖❖❖ EXERCISE-I ❖❖❖

*Modulus, Argument, Conjugate of Complex Numbers SQRT, Cube Root;
Problems on modulus; Problems on Argument*

LEVEL-I (MAIN)

Single answer type questions

- If $\frac{3}{2 + \cos\theta + i\sin\theta} = x + iy$ then $(x-1)(y-3) =$
 - y^2
 - $-y^2$
 - 0
 - 1
- If $(x+iy)^{\frac{1}{3}} = a+ib$ then $\left(\frac{x}{a}\right) + \left(\frac{y}{b}\right) =$
 - $a^2 - b^2$
 - $4(a^2 - b^2)$
 - $2(a^2 - b^2)$
 - $2(a^2 + b^2)$
- If $\frac{\sin(x/2) + \cos(x/2) + i\tan x}{1 + 2i\sin(x/2)}$ is real then $x =$
 - $n\pi$ or $n\pi + \frac{\pi}{4}$
 - $2n\pi$ or $n\pi + \frac{\pi}{4}$
 - $n\pi$ or $n\pi + \frac{\pi}{2}$
 - $n\pi$ or $n\pi + \frac{\pi}{6}$
- If $(1-i)(1-2i)\dots\dots(1-ni) = x-iy$, then 2, 5, 10, ... $(1+n^2) =$
 - $x^2 + y^2$
 - $x + y$
 - $x^2 - y^2$
 - $x - y$
- If $(1 + \cos\theta - i\sin\theta)(1 + \cos 2\theta + i\sin 2\theta) = x + iy$ then $x^2 + y^2 =$
 - $16\cos^2\theta\sin^2\left(\frac{\theta}{2}\right)$
 - $16\sin^2\theta\cos^2\left(\frac{\theta}{2}\right)$
 - $16\sin^2\theta\sin^2\left(\frac{\theta}{2}\right)$
 - $16\cos^2\theta\cos^2\left(\frac{\theta}{2}\right)$
- If $|z_1| = |z_2| = |z_3| = \dots\dots = |z_n| = 1$, then $|z_1 + z_2 + z_3 + \dots\dots + z_n| =$
 - $\left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} + \dots + \frac{1}{z_n}\right|$
 - $\left|\frac{1}{z_1} - \frac{1}{z_2} + \frac{1}{z_3} - \dots + \frac{1}{z_n}\right|$
 - $\left|\frac{1}{z_1^2} + \frac{1}{z_2^2} + \frac{1}{z_3^2} + \dots + \frac{1}{z_n^2}\right|$
 - $\left|\frac{1}{z_1^2} - \frac{1}{z_2^2} + \frac{1}{z_3^2} - \dots + \frac{1}{z_n^2}\right|$
- If z_1, z_2 are two complex numbers satisfying $\left|\frac{z_1 - 3z_2}{3 - z_1z_2}\right| = 1$, $|z_1| \neq 3$, then $|z_2| =$
 - 1
 - 2
 - 3
 - 4

8. $\sqrt{4ab - 2(a^2 - b^2)}i =$

$$1) \pm \left\{ \frac{(a-b) + i(a-b)}{2} \right\}$$

$$2) \pm \{(i + 5b) + (a - 4b)\}$$

$$3) \pm \{(a+b) - i(a-b)\}$$

$$4) \pm \{(i+b) - i(a-b)\}$$

9. $\sqrt{x + i\sqrt{x^4 + x^2 + 1}} =$

$$1) \frac{\sqrt{x^2+x+1} + i\sqrt{x^2-x+1}}{\sqrt{2}}$$

$$2) \frac{\sqrt{x^2+x+1}-i\sqrt{x^2-x+1}}{2}$$

3) $\sqrt{x+1} + i\sqrt{x-1}$

4) $\sqrt{x-1} + i\sqrt{x+1}$

10. If $\sqrt{x+iy} = \pm(a+ib)$ then $\sqrt{-x-iy} =$

$$1) \pm(b+ia)$$

$$2) \pm(a - ib)$$

3) $\pm(a+ib)$

4) $\pm(b-ia)$

11. If z_1 and z_2 are two complex numbers satisfying the equation $\left| \frac{z_1 + z_2}{z_1 - z_2} \right| = 1$, then $\frac{z_1}{z_2}$ is a number which is

1) Positive real

2) Negative real

3) Zero or purely imaginary

4) None of these

12. If $|z_1 - 1| < 1$, $|z_2 - 2| < 2$ and $|z_3 - 3| < 3$, then $|z_1 + z_2 + z_3|$

1) is less than 6

2) is greater than 6

3) is less than 12

4) lies between 6 and 12

13. If z is a complex number such that $-\frac{\pi}{2} \leq \arg Z \leq \frac{\pi}{2}$ then which of the following in equality is true.

$$1) |z - z^*| \leq |z| |\arg z - \arg z^*|$$

$$2) \quad |z - \bar{z}| \leq |\arg z - \arg \bar{z}|$$

$$3) |z - \bar{z}| > |z| |\arg z - \arg \bar{z}|$$

4) $|z - \frac{1}{2}| = 1$

14. If $Z = \begin{vmatrix} 1 & 1-2i & 3+5i \\ 1+2i & -5 & 10i \\ 3-5i & -10i & 11 \end{vmatrix}$ then

1) Z is purely imaginary

2) Z is purely real

3) $Z = 0$
$$4) \arg Z = \frac{-\pi}{4}$$

15. If $z = i^i$ where $i^2 = -1$ then $|z| =$

1) 1

$$2) \ e^{-\pi/2}$$
3) $e^{-\pi}$

4) $e^{\pi/2}$

Numerical value type questions

16. The minimum value of $|z-1|+|z-2|+|z-3|+|z-4|+|z-5|$ is

17. If $Z = i \operatorname{Log}(2 - \sqrt{3})$ then $\cos Z =$
18. If Complex Number Z satisfies the condition $|Z - 2 + i| \leq 1$, then maximum distance of origin from $A(4 + i(2 - z))$ is
19. If $Z = (3 + 7i)(\lambda + \mu i)$ where $\lambda, \mu \in I - \{0\}$ and $i = \sqrt{-1}$ is purely imaginary then minimum value of $|z|^2$ is
20. If $\left|Z - \frac{6}{2}\right| = 5$ and maximum and minimum values of $|z|$ are λ and μ respectively. then $\frac{\lambda^\mu + \mu^\lambda}{\lambda^\mu - \mu^\lambda} =$
21. Number of ordered pairs of (a, b) of real numbers such that $(a + ib)^{2020} = a - ib$ holds good is
22. If z_1, z_2, z_3 are complex numbers such that $|z_1| = |z_2| = |z_3| = 1$ then $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$ can not exceed.
23. If a, b, c, a_1, b_1 and c_1 , are non-zero complex numbers satisfying $\frac{a}{a_1} + \frac{b}{b_1} + \frac{c}{c_1} = 1 + i$ and $\frac{a_1}{a} + \frac{b_1}{b} + \frac{c_1}{c} = 0$ where $i = \sqrt{-1}$, the value of $\frac{\left|\frac{a^2}{a_1^2} + \frac{b^2}{b_1^2} + \frac{c^2}{c_1^2}\right|}{1000} =$

LEVEL-II (ADVANCED)

Single answer type questions

1. If the complex number Z satisfying $Z + |Z| = 2 + 8i$ then value of $|Z| =$
 a) 8 b) 17 c) 15 d) 24
2. If $|Z + 2 - i| = 5$ then maximum value of $|3Z + 9 - 7i| =$
 a) 20 b) 15 c) 5 d) 16
3. If z and w are two complex numbers such that $\bar{z} + i\bar{w} = 0$ and $\arg(zw) = \pi$ then $\arg(z) =$
 a) $\frac{\pi}{4}$ b) $\frac{\pi}{2}$ c) $\frac{3\pi}{4}$ d) $\frac{5\pi}{4}$
4. If Z_1 and Z_2 are two complex numbers such that $Z_1^2 + Z_2^2 \in R$ and $Z_1(Z_1^2 - 3Z_2^2) = 2$;
 $Z_2(3Z_1^2 - Z_2^2) = 11$ then $Z_1^2 + Z_2^2 =$
 a) 5 b) 125 c) 25 d) 15
5. Let $z_r (1 \leq r \leq 4)$ be complex numbers such that $|z_r| = \sqrt{r+1}$ and
 $|30z_1 + 20z_2 + 15z_3 + 12z_4| = k|z_1z_2z_3 + z_2z_3z_4 + z_3z_4z_1 + z_4z_1z_2|$ Then the value of k equals
 a) $|z_1z_2z_3|$ b) $|z_2z_3z_4|$ c) $|z_4z_1z_2|$ d) None of these
6. If $|z_1| = 2, |z_2| = 3, |z_3| = 4$ and $|2z_1 + 3z_2 + 4z_3| = 4$, then absolute value of $8z_2z_3 + 27z_3z_1 + 64z_1z_2$ equals
 a) 24 b) 48 c) 72 d) 96

7. If the ratio $\frac{1-z}{1+z}$ is purely imaginary, then
 a) $0 < |z| < 1$ b) $|z| = 1$
 c) $|z| > 1$ d) bounds for $|z|$ can not be decided
8. If P and Q are represented by the numbers z_1 and z_2 such that $\left| \frac{1}{z_2} + \frac{1}{z_1} \right| = \left| \frac{1}{z_2} - \frac{1}{z_1} \right|$, then the circumcentre of ΔOPQ , (where O is the origin) is
 a) $\frac{z_1 - z_2}{2}$ b) $\frac{z_1 + z_2}{2}$ c) $\frac{z_1 + z_2}{3}$ d) $z_1 + z_2$
9. If z_1, z_2 are two complex numbers satisfying the equation $\left| \frac{z_1 + z_2}{z_1 - z_2} \right| = 1$, then $\frac{z_1}{z_2}$ is a number which is
 a) Positive real b) Negative real
 c) Zero d) Lying on imaginary axis
10. Let z be a complex number satisfying $|z + 16| = 4|z + 1|$ then
 a) $|z| = 4$ b) $|z| = 5$ c) $|z| = 6$ d) $3 < |z| < 6$
11. If $c^2 + s^2 = 1$, then $\frac{1 + c + is}{1 + c - is}$ is equal to
 a) $c - is$ b) $c + is$ c) $s + ic$ d) $s - ic$
12. If $\log_{\tan 30^\circ} \left(\frac{2|z|^2 + 2|z| - 3}{|z| + 1} \right) < -2$ then
 a) $|z| < \frac{3}{2}$ b) $|z| > \frac{3}{2}$ c) $|z| > 2$ d) $|z| < 2$
13. If $|z| = \min\{|z-1|, |z+1|\}$ then $|z + \bar{z}| =$
 a) 1 b) 2 c) 3 d) 4
14. If z_1 is a root of the equation $a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n = 3$, where $|a_i| < 2$ for $i = 0, 1, \dots, n$. Then
 a) $|z_1| > \frac{1}{3}$ b) $|z_1| < \frac{1}{4}$ c) $|z_1| > \frac{1}{4}$ d) $|z_1| < \frac{1}{3}$
15. If $(a + ib)^5 = \alpha + i\beta$ then $(b + ia)^5$ is equal to
 a) $\beta - i\alpha$ b) $\beta + i\alpha$ c) $\alpha - \beta$ d) $-\alpha - i$
16. If $(1 + i)(1 + 2i)(1 + 3i) \dots (1 + ni) = \alpha + i\beta$, then $2 \cdot 5 \cdot 10 \dots (1 + n^2)$ is equal to (where $\alpha, \beta, n \in \mathbb{R}$)
 a) $\alpha - i\beta$ b) $\alpha^2 - \beta^2$ c) $\alpha^2 + \beta^2$ d) none of these
17. If $z = (1 + \sqrt{3}i)^{10} + (1 - \sqrt{3}i)^{10}$, then $\arg z$ is
 a) $\frac{\pi}{2}$ b) π c) $\frac{\pi}{4}$ d) none of these
18. If Z_1, Z_2 are two complex numbers such that $|Z_1| = 1, |Z_2| = 1$ then the maximum value of $|Z_1 + Z_2| + |Z_1 - Z_2|$ is
 a) 2 b) $2\sqrt{2}$ c) 4 d) none of these

19. Let z be a complex number satisfying $|z+16|=4|z+1|$ then
 a) $|z|=4$ b) $|z|=5$ c) $|z|=6$ d) $3<|z|<6$
20. If (a, b) lies on $x^2 + y^2 = 100$ and z_1, z_2 lies on curve $z^2 + \bar{z}^2 = 12$ then $\frac{|az_1 - bz_2|^2 + |bz_1 + az_2|^2}{10(|z_1|^2 + |z_2|^2)} =$
 a) 1 b) 10 c) 100 d) $\sqrt{10}$
21. If $|z|=1$ and $w = \frac{z-1}{z+1}$, ($z \neq -1$) then $\text{Re}(w)$ is
 a) 0 b) $\frac{1}{|z+1|^2}$ c) $\frac{z}{|z+1|} \frac{1}{|z+1|^2}$ d) $\frac{\sqrt{2}}{|z+1|^2}$
22. The least value of $n \in \mathbb{N}$ such that $(1+i)^n \pi = (1-i)^n 2 \sin^{-1} \left(\frac{1+x^2}{2x} \right)$
 a) 2 b) 4 c) 8 d) 16

More than one correct answer type questions

23. If a and b are complex numbers, then $|a + \sqrt{a^2 - b^2}| + |a - \sqrt{a^2 - b^2}|$ is
 a) Equal to $|a+b| + |a-b|$ b) Equal to $|a+b| - |a-b|$
 c) $\leq \frac{1}{2}(|a|^2 + |b|^2)$ d) $\leq 2(|a| + |b|)$
24. If $z = \sqrt{20i-21} + \sqrt{21-20i}$, then the principal value of $\arg 'z'$ can be
 a) $\frac{\pi}{4}$ b) $\frac{3\pi}{4}$ c) $-\frac{3\pi}{4}$ d) $-\frac{\pi}{4}$
25. If $\arg(z^{3/8}) = (1/2)\arg(z^2 + \bar{z}^{1/2})$ then which of the following is (are) true ?
 a) $|z|=1$ b) z is real c) z is pure imaginary d) $z^{1/2} = 1$
26. For any complex number $z = x + iy$, define $(z) = |x| + |y|$. If z_1 and z_2 are any complex numbers, then
 a) $(z_1 + z_2) \leq (z_1) + (z_2)$ b) $(z_1 + z_2) = (z_1) + (z_2)$
 c) $(z_1 + z_2) \geq (z_1) + (z_2)$ d) $|(z_1 + z_2)| \leq |(z_1)| + |(z_2)|$
27. If for any 2 complex nos. z_1 and z_2 , ($|z_2| \neq 1$) $\sqrt{z_1} - i\sqrt{z_2} = |z_2| \sqrt{z_1} + i|z_1| \sqrt{z_2}$, then which of the following options are correct
 a) $z_1 \bar{z}_2 + 1 = 0$ b) $z_1 + z_2 = 0$ c) $\arg\left(\frac{z_1}{z_2}\right) = \pi$ d) $\arg\left(\frac{z_1}{z_2}\right) = \frac{\pi}{2}$
28. Let z satisfies $\arg(z - 3 - 4i) = \arg(z - 1 + 2i) + \arg(iz) + \arg(\bar{z})$ and z_1 is a fixed complex number such that $|z - z_1|$ is constant then
 a) z lies on a semi-circle b) $|z_1^8|$ is equal to 625
 c) argument of z_1 is $\frac{\pi}{4}$ d) arg of $(z_1 - 2)$ is $\frac{\pi}{2}$

29. For points z_1, z_2, z_3, z_4 in complex plane such that $|z_1| < 1, |z_2| = 1$ and $|z_3| \leq 1$ and $z_3 = \frac{z_2(z_1 - z_4)}{\bar{z}_1 z_4 - 1}$, then $|z_4|$ can be
- a) 2 b) $\frac{2}{5}$ c) $\frac{1}{3}$ d) $\frac{5}{2}$
30. If z_1 lies on $|z| = 1$ and z_2 lies on $|z| = 2$, then
- a) $3 \leq |z_1 - 2z_2| \leq 5$ b) $1 \leq |z_1 + z_2| \leq 3$ c) $|z_1 - 3z_2| \geq 5$ d) $|z_1 - z_2| \geq 1$
31. If $z_1 = a + ib; z_2 = c + id$ are complex numbers such that $|z_1| = |z_2| = 1$ and $\operatorname{Re}(z_1 \bar{z}_2) = 0$, then the pair of complex numbers $\omega_1 = a + ic$ and $\omega_2 = b + id$ satisfies
- a) $|\omega_1| = 1$ b) $|\omega_2| = 1$ c) $\operatorname{Re}(\omega_1 \bar{\omega}_2) = 0$ d) $\operatorname{Re}(\omega_1 \bar{\omega}_2) = 1$

Linked comprehension type questions

Passage - I :

Let α, β, γ be three real numbers such that $\alpha^2 + \beta^2 + \gamma^2 - \gamma = 0$ and $z = \frac{\alpha + i\beta}{1 - \gamma}$

32. $|z|^2$ equals
- a) γ b) $1 - \gamma$ c) $\frac{\gamma}{1 - \gamma}$ d) $\frac{1 - \gamma}{\gamma}$
33. α equals
- a) $\frac{z + \bar{z}}{2(1 + |z|^2)}$ b) $\frac{(\bar{z} - z)i}{2(1 + |z|^2)}$ c) $\frac{(z - \bar{z})i}{2(1 + |z|^2)}$ d) $\frac{(z - \bar{z})i}{2(1 + |z|^2)}$
34. β equals
- a) $\frac{z - \bar{z}}{2(1 + |z|^2)}$ b) $\frac{(\bar{z} - z)i}{2(1 + |z|^2)}$ c) $\frac{z + \bar{z}}{2(1 + |z|^2)}$ d) $\frac{(z - \bar{z})i}{2(1 + |z|^2)}$

Passage-II :

Let Z is any complex number satisfying the equation $z^2 + pz + q = 0$ ($q \in \mathbb{C}$) and both the roots of the equation have unit modulus

35. Modulus of q is
- a) $\frac{1}{2}$ b) 2 c) 1 d) 3
36. Which of following is true
- a) $|p| \leq 2$ b) $|p| \leq 1$ c) $|p| \geq 1$ d) $|p| \geq 2$
37. If $\arg(p) = \alpha$ and $\arg(q) = \beta$, then relation between α and β is
- a) $\alpha = 2\beta$ b) $2\alpha = \beta$ c) $\alpha = \frac{1}{\beta}$ d) $\alpha = -\beta$

Passage-III :

Given $z^{-1} = (a + ib)^{-1} + (a + ic)^{-1}$, $z = x + iy, a, b, c$ being real with $a + ib, a + ic$ not zero.

38. The value of $x^2 + y^2$ equals

- a) $\frac{(a^2 - b^2)(a^2 - c^2)}{4a^2 + (b+c)^2}$ b) $\frac{(a^2 + b^2)(a^2 + c^2)}{4a^2 + (b+c)^2}$ c) $\frac{(a^2 + c^2)(a+b)}{a^2 + b^2}$ d) $a^2 + b^2 + c^2$

39. The value of $(x-a)^2 + y^2$ equals

- a) $\frac{(a^2 - bc)^2}{4a^2 + (b+c)^2}$ b) $\frac{a^2}{4a^2 + c^2}$ c) $\frac{a^2 + b^2 + c^2}{a^2 + b^2 + 4c^2}$ d) $\frac{(a^2 + bc)^2}{4a^2 + (b+c)^2}$

40. The value of $\operatorname{Re}(z)$ i.e., 'x' equals

- a) $\frac{2a^2 + b^2 + c^2}{4a^2 + (b+c)^2}$ b) $\frac{a(2a^2 + b^2 + c^2)}{4a^2 + b^2}$ c) $\frac{2(2a^2 + b^2 + c^2)}{a^2 + b^2 + c^2}$ d) $\frac{a(2a^2 + b^2 + c^2)}{4a^2 + (b+c)^2}$

Passage-IV :

If P is the point (a, b) on the Argand plane corresponding to the complex number $Z = a + ib$, then

$\overrightarrow{OP} = a\hat{i} + b\hat{j}$. $\therefore |\overrightarrow{OP}| = \sqrt{a^2 + b^2} = |Z|$ and $\arg(Z) = \text{direction of the vector } \overrightarrow{OP} = \tan^{-1}\left(\frac{b}{a}\right)$.

Therefore, complex number Z can also be represented by $\overrightarrow{OP}$.

41. Let (r, θ) denote the point $r(\cos\theta + i\sin\theta)$ in the Argand plane. If $a = (1, \alpha)$, $b = (1, \beta)$, $c = (1, \gamma)$ and $a + b + c = 0$ then $a^{-1} + b^{-1} + c^{-1} =$

- a) 1 b) -1 c) 0 d) None of these

42. Let Z_1, Z_2 be two complex numbers such that $|Z_1 + Z_2| = |Z_1| + |Z_2|$, then

- a) $\arg(Z_1) = \arg(Z_2)$ b) $\arg(Z_1) + \arg(Z_2) = 0$
c) $\arg(Z_2/Z_1) = 1$ d) None of these

43. If Z_1, Z_2 and Z_3, Z_4 are two pairs of conjugate complex numbers, then $\arg\left(\frac{Z_1}{Z_4}\right) + \arg\left(\frac{Z_2}{Z_3}\right) =$

- a) 0 b) $\frac{\pi}{2}$ c) $\frac{3\pi}{2}$ d) π

Matrix matching type questions

44. If $z = \cos\alpha + i\sin\alpha, 0 < \alpha < \pi/6$, then the principal argument of:

COLUMN - I

COLUMN - II

A) $1 + z^3$

p) $2\alpha - \frac{\pi}{2}$

B) $1 - z^4$

q) $\frac{\pi}{2} + \frac{\alpha}{2}$

C) $\frac{1+z^3}{1-z^4}$

r) $\frac{3\alpha}{2}$

D) $\frac{z^4 - 1}{z^3 + 1}$

s) $\frac{\pi}{2} - \frac{\alpha}{2}$

45. COLUMN - I

COLUMN - II

A) If $Z_1 = 1+i, Z_2 = \sqrt{3}+i$, then $\arg\left(\frac{Z_1}{Z_2}\right) =$

p) $\frac{3\pi}{4}$

B) If $|Z-i| = \sqrt{2}$. Then $\arg\left(\frac{Z-1}{Z+1}\right) =$

q) $\pm \frac{\pi}{2}$

C) If $\bar{Z} + i\bar{W} = 0$, $\arg(Z.W) = \pi$, then $\arg(Z) =$

r) $\frac{\pi}{3}$

D) If $|Z^2 - 1| = |Z^2| + 1$, then $\arg(Z) =$

s) $\frac{\pi}{4}$

Integer answer type questions

46. If $z \neq 0$ and $2 + \cos\theta + i\sin\theta = \frac{3}{z}$, then the value of $2(z + \bar{z}) - |z|^2$ is

47. The number of complex numbers z satisfying the conditions $\left|\left(\frac{z}{\bar{z}}\right) + \left(\frac{\bar{z}}{z}\right)\right| = 1, |z| = 1$ and $\arg z \in (0, 2\pi)$ is

48. If $a, b \in \mathbb{R}$, then the number of complex numbers $a + ib$ for which $(a+ib)^2 = (a-ib)^3$ is

49. Number of solution of the equation $z^3 = \bar{z}i|z|$ are _____

50. If α, β, γ and a, b, c are complex numbers such that $\frac{\alpha}{a} + \frac{\beta}{b} + \frac{\gamma}{c} = 1+i$ and $\frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma} = 0$ then the value of $\left|\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2} + \frac{\gamma^2}{c^2}\right|$ is equal to

51. The quadratic equation $z^2 + (a+ib)z + (c+id) = 0$ (a, b, c, d are real and $bd \neq 0$) has equal roots. Then the value of ab/d is _____

EXERCISE-II

Demoivers theorem, n th roots of unity etc.,

LEVEL-I (MAIN)

Single answer type questions

1. $[(\cos\alpha - \cos\beta) + i(\sin\alpha - \sin\beta)]^n + [(\cos\alpha - \cos\beta) - i(\sin\alpha - \sin\beta)]^n =$

1) $2^{n+1} \cdot \sin^n\left(\frac{\alpha-\beta}{2}\right) \cos\left[n\left(\frac{\pi}{2} + \frac{\alpha+\beta}{2}\right)\right]$

2) $2^{n+1} \cdot \sin^n\left(\frac{\alpha-\beta}{2}\right) \cos\left[n\left(\frac{\pi}{2} + \frac{\alpha+\beta}{2}\right)\right]$

3) $2^{n+1} \cdot \sin^n\left(\frac{\alpha+\beta}{2}\right) \cos\left[n\left(\frac{\pi}{2} + \frac{\alpha-\beta}{2}\right)\right]$

4) $2^{n+1} \cdot \sin^n\left(\frac{\alpha+\beta}{2}\right) \cos\left[n\left(\frac{\pi}{2} + \frac{\alpha-\beta}{2}\right)\right]$

2. If $x = \cos\alpha + i\sin\alpha$, $y = \cos\beta + i\sin\beta$, then $\frac{x}{y} - \frac{y}{x} =$

1) $2i\sin(\alpha+\beta)$

2) $2i\sin(\alpha-\beta)$

3) $2i\cos(\alpha+\beta)$

4) $2i\cos(\alpha-\beta)$

3. If n is a positive integer and $(1+i\sqrt{3})^n + (1-i\sqrt{3})^n = 2^{n+1} \cos \theta$, then the value of θ is
 - 1) $n\pi/3$
 - 2) $n\pi/2$
 - 3) $n\pi/4$
 - 4) $n\pi/6$
4. If n is a positive integer, then the $(a+ib)^{m/n} + (a-ib)^{m/n} =$
 - 1) $2(a^2+b^2)^{m/2n} \cos \left[\frac{m}{n} \tan^{-1} \frac{b}{a} \right]$
 - 2) $2(a^2+b^2)^{m/2n} \sin \left[\frac{m}{n} \tan^{-1} \frac{b}{a} \right]$
 - 3) $2(a^2+b^2)^{m/2n} \cot \left[\frac{m}{n} \tan^{-1} \frac{b}{a} \right]$
 - 4) $2(a^2+b^2)^{m/2n} \tan \left[\frac{m}{n} \tan^{-1} \frac{b}{a} \right]$
5. The product of the four values of $\left(\frac{1+i\sqrt{3}}{2} \right)^{3/4}$ is
 - 1) -1
 - 2) 1
 - 3) i
 - 4) $-i$
6. If $(\sqrt{3}+i)^n = 2^n$, $n \in I$, the set of integers, then n is a multiple of
 - 1) 6
 - 2) 10
 - 3) 9
 - 4) 12
7. If $x_n = \cos \frac{\pi}{2^n} + i \sin \frac{\pi}{2^n}$, then $\prod_{n=1}^{\infty} x_n =$
 - 1) -1
 - 2) 1
 - 3) $1/\sqrt{2}$
 - 4) $i/\sqrt{2}$
8. The value of $1 + \sum_{k=0}^{14} \left\{ \cos \frac{(2k+1)\pi}{15} + i \sin \frac{(2k+1)\pi}{15} \right\}$ is
 - 1) 0
 - 2) 1
 - 3) -1
 - 4) i
9. If $\cos \alpha + \cos \beta + \cos \gamma = 0 = \sin \alpha + \sin \beta + \sin \gamma$ then $x^{\sin(2\alpha-\beta-\gamma)} \cdot x^{\sin(2\beta-\gamma-\alpha)} \cdot x^{\sin(2\gamma-\alpha-\beta)} =$
 - 1) 0
 - 2) 1
 - 3) x
 - 4) x^3
10. If $A+B+C = \pi$ and $\cos A + \cos B + \cos C = 0 = \sin A + \sin B + \sin C$ then $\cos 3A + \cos 3B + \cos 3C =$
 - 1) 3
 - 2) -3
 - 3) 0
 - 4) 1
11. The common roots of the equation $x^{12} - 1 = 0$, $x^4 + x^2 + 1 = 0$ are
 - 1) ω, ω^2
 - 2) $-\omega, -\omega^2$
 - 3) $\pm\omega, \pm\omega^2$
 - 4) $\pm\omega^2, \pm\omega^5$
12. The product of all n^{th} roots of unity ($n > 1$) is
 - 1) 0
 - 2) $(-1)^{n-1}$
 - 3) -1
 - 4) 1
13. If $1, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are the n^{th} roots of unity then $(1-\alpha_1)(1-\alpha_2)\dots(1-\alpha_{n-1}) =$
 - 1) $n-1$
 - 2) n
 - 3) -1
 - 4) 1
14. If $1, \alpha_1, \alpha_2, \dots, \alpha_{n-1}$ are the n^{th} roots of unity and n is an odd natural number then $(1+\alpha_1)(1+\alpha_2)\dots(1+\alpha_{n-1}) =$
 - 1) 1
 - 2) -1
 - 3) 0
 - 4) -2

15. If α is an n^{th} root of unity, the $1 + 2\alpha + 3\alpha^2 + \dots + n\alpha^{n-1} =$
- 1) $\frac{n}{1-\alpha}$ 2) $\frac{-n}{1-\alpha}$ 3) $\frac{-n}{(1-\alpha)^2}$ 4) $\frac{n}{(1-\alpha)^2}$
16. If $x^2 + x + 1 = 0$, then the value of $\left(x + \frac{1}{x}\right)^2 + \left(x^2 + \frac{1}{x^2}\right)^2 + \dots + \left(x^{27} + \frac{1}{x^{27}}\right)^2$ is
- 1) 27 2) 72 3) 45 4) 54
17. If ω is a complex cube root of unity, then the value of $\frac{a+b\omega+c\omega^2}{c+a\omega+b\omega^2} + \frac{a+b\omega+c\omega^2}{b+c\omega+a\omega^2}$ is
- 1) 0 2) -1 3) 1 4) 2
18. If ω is an imaginary cube root of unity, then the value of the expression $1(2-\omega)(2-\omega^2) + 2(3-\omega)(3-\omega^2) + \dots + (n-1)(n-\omega)(n-\omega^2)$ is
- 1) $\frac{1}{4}n^2(n+1)^2 - n$ 2) $\frac{1}{4}n^2(n+1)^2 + n$ 3) $\frac{1}{4}n^2(n+1) - n$ 4) $\frac{1}{4}n(n+1)^2 - n$
19. If $\omega \neq 1$ is a cube root of unity satisfying $\frac{1}{a+w} + \frac{1}{b+w} + \frac{1}{c+w} = 2w^2$ and $\frac{1}{a+w^2} + \frac{1}{b+w^2} + \frac{1}{c+w^2} = 2w$ then the value of $\frac{1}{a+1} + \frac{1}{b+1} + \frac{1}{c+1}$ is
- 1) 2 2) -2 3) ω^2 4) ω
20. If z is a complex number satisfying $z^4 + z^3 + 2z^2 + z + 1 = 0$, then the set of possible values of $|z|$ is
- 1) {1, 2} 2) {1} 3) {1, 2, 3} 4) {1, 2, 3, 4}
21. If $|z|=1$ then $\left(\frac{1+z}{1+\bar{z}}\right)^n + \left(\frac{1+\bar{z}}{1+z}\right)^n$ is equal to
- 1) $2\cos(n(\arg z))$ 2) $2\sin(\arg z)$ 3) $2\cos\left(n\left(\arg\left(\frac{z}{2}\right)\right)\right)$ 4) $2\sin\left(n\left(\arg\left(\frac{z}{2}\right)\right)\right)$
22. If $\omega \neq 1$ be a cube root of unity and $(1+\omega)^7 = l + m\omega$, then the value of $l + m =$
- 1) 0 2) 1 3) 2 4) -1
23. If $2w+1 = \sqrt{3}i$, then the determinant $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1-w^2 & w^2 \\ 1 & w^2 & w^4 \end{vmatrix}$ is
- 1) $3w$ 2) $3w(w-1)$ 3) $3w^2$ 4) $3w(1-w)$
24. If w is complex cube root of 1, then $\cos\left(\left((1-w)(1-w^3) + (2-w)(2-w^2) + \dots + (10-w)(10-w^2)\right)\frac{\pi}{900}\right) =$
- 1) -1 2) 0 3) 1 4) $\frac{\sqrt{3}}{2}$

25. $S_1 \equiv$ If $z^2 + z + 1 = 0$ and $n \in \mathbb{N}$ then $\sum_{k=1}^n z^k + z^{-k} = -n + 3 \left[\frac{n}{3} \right]$ where $[]$ is
 $S_2 \equiv$ If $w \neq 1$ is a cube root of 1 then $w^k + w^{-k} = \begin{cases} -1 & \text{if } k \text{ is not a multiple of } 3 \\ 2 & \text{if } k \text{ is a multiple of } 3 \end{cases}$ then
 1) S_1, S_2 both true and S_2 explains S_1
 2) S_1, S_2 both true but S_2 is not a correct explanation of S_1
 3) S_1 true, S_2 False
 4) S_1 false, S_2 true
26. The area of the figure formed by the roots of $z^5 = (z - 1)^5$ in argand plane is
 1) $32 \cos \frac{2\pi}{5}$ 2) $1 - \cos^2 \frac{2\pi}{5} - \sin^2 \frac{2\pi}{5}$ 3) $\cot \frac{\pi}{5}$ 4) $2 \cot \frac{\pi}{5}$
27. If α is non real fifth root of 1 then $\log_2 \left| 1 + \alpha + \alpha^2 + \alpha^3 - \frac{1}{\alpha} \right|$
 1) 0 2) $\frac{1}{2}$ 3) 1 4) 2
28. If 1, $\alpha_1, \alpha_2, \dots, \alpha_{99}$ are roots of $Z^{100} = 1$ then $\sum_{1 \leq i < j \leq 99} \alpha_i \alpha_j$ equal to
 1) 0 2) 1 3) 99 4) 100
29. If $\cos x + 2 \cos y + 3 \cos z = \sin x + 2 \sin y + 3 \sin z = 0$ then the value of $\sin 3x + 8 \sin 3y + 27 \sin 3z$ is
 1) $\sin(x+y+z)$ 2) $3 \sin(x+y+z)$ 3) $18 \sin(x+y+z)$ 4) $\sin(x+2y+z)$

Numerical value type questions

30. If $Z + \frac{1}{Z} = 2 \cos 6^\circ$ and $z^{1000} + \frac{1}{z^{1000}} + 1$ is
 31. If $(\sqrt{3} + i)^{100} = 2^{99}(a + ib)$ then $\frac{a^2 + b^2}{100} =$
 32. If n is not a multiple of 3, and if $\lambda = \sin \left\{ (\omega^n + \omega^{2n})\pi - \frac{\pi}{4} \right\}$ then $\frac{\lambda}{\sqrt{2}} =$
 33. If $a = 1 + \frac{1}{3} + \frac{1}{6} + \frac{1}{9} + \dots$, $b = 1 + \frac{1}{4} + \frac{1}{7} + \dots$, $c = \frac{1}{2} + \frac{1}{5} + \frac{1}{8} + \dots$ then $a^3 + b^3 + c^3 - 3abc =$
 34. If m and n are the smallest positive integers satisfying the relation $\left(2 \operatorname{cis} \frac{\pi}{6} \right)^m = \left(4 \operatorname{cis} \frac{\pi}{4} \right)^n$ where $i = \sqrt{-1}$ $\left(\frac{n}{m} \right)^2$ equals to

LEVEL-II (ADVANCED)

Single answer type questions

1. α, β, γ are the roots of $x^3 - 3x^2 + 3x + 7 = 0$ (ω is cube root of unity) then $\left(\frac{\alpha-1}{\beta-1} + \frac{\beta-1}{\gamma-1} + \frac{\gamma-1}{\alpha-1} \right)$ is
 a) $\frac{3}{\omega}$ b) ω^2 c) $2\omega^2$ d) 3ω

2. If α, β, γ are the cube roots of $p (< 0)$, then $\frac{x\alpha + y\beta + z\gamma}{x\beta + y\gamma + z\alpha}$ for any x, y, z is equal to (where ω is complex cube root of unity)
- a) 1 b) 0 c) ω^2 d) 3
3. The number of common roots of the equations $x^3 + 2x^2 + 2x + 1 = 0$ and $x^{2012} + x^{2014} + 1 = 0$ is
- a) 1 b) 2 c) 3 d) 4
4. If $\alpha = -1 + i\sqrt{3}$ and n is a positive integer which is not a multiple of 3, then $\alpha^{2n} + 2^n \alpha^n + 2^{2n} =$
- a) 1 b) -1 c) 0 d) α^2
5. If z is a non real root of $\sqrt[7]{-1}$ then $z^{86} + z^{175} + z^{289}$ is equal to
- a) 0 b) -1 c) 3 d) 1
6. Number of ordered pairs of (a, b) of real numbers such that $(a + ib)^{2008} = a - ib$ holds good, is
- a) 2008 b) 2009 c) 2010 d) 1
7. Let $f(x) = x^4 + ax^3 + bx^2 + cx + d$, $a, b, c, d \in \mathbb{R}$. Suppose all roots of $f(x) = 0$ are real and $|f(i)| = 1$, then $a + b^2 + c^3 + d^4$ equals
- a) -1 b) 1 c) 4 d) None of the above
8. All the roots of the equation $11z^{10} + 10iz^9 + 10iz - 11 = 0$ lie
- a) inside $|z| = 1$ b) on $|z| = 1$ c) outside $|z| = 1$ d) can't say
9. Let $z = e^{i\theta}$ then the value of $\sum_{m=1}^{15} \operatorname{Im}(z^{2m-1})$, when $\theta = 2^\circ$ is
- a) $\frac{1}{\sin 2^\circ}$ b) $\frac{1}{3\sin 2^\circ}$ c) $\frac{1}{2\sin 2^\circ}$ d) $\frac{1}{4\sin 2^\circ}$
10. If $az^2 + bz + 1 = 0$, $a, b, z \in \mathbb{C}$ and $|a| = \frac{1}{2}$, have a root α such that $|\alpha| = 1$ then $|a\bar{b} - b| =$
- a) 1/4 b) 1/2 c) 5/4 d) 3/4
11. Let $C = \cos \frac{2\pi}{7} + \cos \frac{4\pi}{7} + \cos \frac{8\pi}{7}$ and $S = \sin \frac{2\pi}{7} + \sin \frac{4\pi}{7} + \sin \frac{8\pi}{7}$, then
- a) $C = \frac{1}{2}$ b) $S = \frac{1}{2}$ c) $C = \frac{\sqrt{7}}{2}$ d) $S = \frac{\sqrt{7}}{2}$

More than one correct answer type questions

12. If $S = \sum_{k=1}^{10} \left(\sin \frac{2\pi k}{11} - i \cos \frac{2\pi k}{11} \right)$ then

- a) $S + \bar{S} = 0$ b) $S\bar{S} = 1$ c) $\sqrt{S} = \pm \frac{1}{\sqrt{2}}(1+i)$ d) $S - \bar{S} = 0$

13. Let $P(x)$ and $Q(x)$ be two real polynomials. Suppose that $f(x) = P(x^3) + xQ(x^3)$ is divisible by $x^2 + x + 1$, then
- $P(x)$ is divisible by $(x - 1)$ but $Q(x)$ is not divisible by $x - 1$
 - $Q(x)$ is divisible by $(x - 1)$ but $P(x)$ is not divisible by $x - 1$
 - Both $P(x)$ and $Q(x)$ are divisible by $x - 1$
 - $f(x)$ is divisible by $x - 1$
14. Let a complex number $\alpha, (\alpha \neq 1)$ to be root of the equation $Z^{p+q} - Z^p - Z^q + 1 = 0$, where p, q are distinct primes. Then
- $1 + \alpha + \alpha^2 + \dots + \alpha^{p-1} = 0$ and $1 + \alpha + \alpha^2 + \dots + \alpha^{q-1} \neq 0$
 - $1 + \alpha + \alpha^2 + \dots + \alpha^{p-1} \neq 0$ and $1 + \alpha + \alpha^2 + \dots + \alpha^{q-1} = 0$
 - $1 + \alpha + \alpha^2 + \dots + \alpha^{p-1} = 0$ and $1 + \alpha + \alpha^2 + \dots + \alpha^{q-1} = 0$
 - $1 + \alpha + \alpha^2 + \dots + \alpha^{p-1} \neq 0$ and $1 + \alpha + \alpha^2 + \dots + \alpha^{q-1} \neq 0$
15. If $(1+x)^n = C_0 + C_1x + \dots + C_nx^n$, where n is a positive integer, then
- $C_0 - C_2 + C_4 - \dots = 2^{n/2} \cos\left(\frac{n\pi}{4}\right)$
 - $C_1 - C_3 + C_5 - \dots = 2^{n/2} \sin\left(\frac{n\pi}{4}\right)$
 - $C_0 + C_4 + C_8 + \dots = 2^{n-2} + 2^{\left(\frac{n-2}{2}\right)} \cos\left(\frac{n\pi}{4}\right)$
 - $C_0 + C_3 + C_6 + \dots = \frac{1}{3} \left(2^n + 2 \cos\left(\frac{n\pi}{3}\right) \right)$

Linked comprehension type questions

Passage-I :

If n is a natural number, define polynomial $P_n(x)$ of degree n as follows:

$$\cos n\theta = P_n(\cos \theta)$$

18. The value of $\sum_{r=1}^{n-1} \frac{1}{(2-\alpha^r)}$ is equal to
- a) $(n-2)2^n$ b) $\frac{(n-2)2^{n-1}+1}{2^n-1}$ c) $\frac{(n-2)2^{n-1}}{2^n-1}$ d) $\frac{(n-1)2^{n-1}}{2^n-1}$
19. The algebraic sum of perpendicular distance from the points $1, \alpha, \alpha^2, \dots, \alpha^{n-1}$ to the line $\bar{a}z + a\bar{z} + b = 0$ (where α is complex number and 'b' is real) is equal to
- a) $\frac{n}{2|a|}$ b) $\frac{n|b|}{2a}$ c) $\frac{nb}{|a|}$ d) $\frac{nb}{2|a|}$
20. Given $Z = \cos\left(\frac{2\pi}{2n+1}\right) + i \sin\left(\frac{2\pi}{2n+1}\right)$, n is a positive integer, then the equation whose roots are $\alpha = Z + Z^3 + Z^5 + \dots + Z^{2n-1}$ and $\beta = Z^2 + Z^4 + \dots + Z^{2n}$ is
- a) $x^2 + x + 1 = 0$ b) $x^2 + x - 1 = 0$
- c) $x^2 + x + \frac{1}{4 \cos^2 \frac{\pi}{2n+1}} = 0$ d) $x^2 - x + \frac{1}{4 \cos^2 \frac{\pi}{2n+1}} = 0$

Passage-III :

Let $A_1, A_2, A_3, \dots, A_n$ be vertices of a regular polygon of n sides whose centre 'O' is z_0 let $A_1, A_2, A_3, \dots, A_n$ be $z_1, z_2, \dots, z_n$ respectively. Let $OA_1 = OA_2 = \dots = OA_n = 1$

21. The value of $|z_1|^2 + |z_2|^2 + |z_3|^2 + \dots + |z_n|^2 = k|z_0|^2$ then k
- a) 1 b) 3 c) n d) n^2
22. If $Z_0 = (0,0)$, $|A_1 A_2|^2 + |A_1 A_3|^2 + \dots + |A_1 A_n|^2 = k|A_1 A_2||A_1 A_3| \dots |A_1 A_n|$ then $k =$
- a) 0 b) 1 c) 2 d) $2n$

Matrix matching type questions**23. COLUMN - I****COLUMN - II**

A) If $|a_i| < 1; \lambda_i \geq 0$ for $i = 1, 2, 3, \dots, n$ and

p) $|z|^n + \frac{1}{|z|}$

$\lambda_1 + \lambda_2 + \lambda_3 + \dots + \lambda_n = 1$ and ω is a

complex cube root of unity, then

$|\lambda_1 a_1 \omega + \lambda_2 a_2 \omega^2 + \dots + \lambda_n a_n \omega^n|$ cannot exceed

B) If $\operatorname{Re}(z) < 0$, then the value of $(1+z+z^2+\dots+z^n)$ cannot exceed

q) 2

C) If $\omega (\neq 1)$ is a cube root of unity, then

r) n

$\frac{1}{\sqrt{3}} |1 + 2\omega + 3\omega^2 + \dots + 3n\omega^{3n-1}|$ ($n \in N$) cannot exceed

D) If $1, \omega, \omega^2, \dots, \omega^{n-1}$ are the n , n^{th} roots of unity, then

s) $2^n - 1$

$(2-\omega)(2-\omega^2) \dots (2-\omega^{n-1}) =$

t) $|a_1| + |a_2| + \dots + |a_n|$

a) A - q ; B - t ; C - s ; D - r

b) A - q ; B - r ; C - s ; D - q

c) A - t ; B - p ; C - r ; D - s

d) A - s ; B - qr ; C - qt ; D - t

24. If z_1, z_2, z_3, z_4 are the roots of the equation $z^4 + z^3 + z^2 + z + 1 = 0$, then

COLUMN - I

COLUMN - II

A) $\left| \sum_{i=1}^4 z_i^4 \right|$ is equal to

p) 0

B) $\sum_{i=1}^4 z_i^5$ is equal to

q) 4

C) $\prod_{i=1}^4 (z_i + 2)$ is equal to

r) 1

D) Least value of $[z_1 + z_2]$ is

s) 11

where $[]$ represents greatest integer function

t) $\left| 4 \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right) \right|$

25. COLUMN - I

COLUMN - II

A) Let $z_k (k=0,1,2,3,4,5,6)$ be the roots of the equation

p) -1

$(z+1)^7 + (z)^7 = 0$ then $\sum_{k=0}^6 \operatorname{Re}(z_k)$

B) If α, β, γ and a, b, c are complex numbers such that

q) $-\frac{7}{2}$

$\frac{\alpha}{a}, \frac{\beta}{b}, \frac{\gamma}{c} = 1+i$ and $\frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma} = 0$, then the value of $\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2} + \frac{\gamma^2}{c^2}$

C) If $z_1, z_2, \dots, z_6$ are six roots of the equations

r) 1

$z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = 0$, then the value of $\prod_{i=1}^6 (z_i + 1)$

D) If $|z| = \min(|z-1|, |z+1|)$, then the value of $|z + \bar{z}|$

s) 4

Integer answer type questions

26. If $a_1, a_2, \dots, a_n$ are real numbers and $\cos \alpha + i \sin \alpha$ is a root of $x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n = 0$ then the value of $|a_1 \cos \alpha + a_2 \cos 2\alpha + \dots + a_n \cos(n\alpha)|$ is

27. If $z = \frac{1}{2}(\sqrt{3} - i)$ and the smallest value of the positive integer n for which $(z^{89} + i^{97})^{94} = z^n$ is λ then $\frac{\lambda}{2} =$

28. If $1, x_1, x_2, x_3$ are the roots of $x^4 - 1 = 0$ and ω is a complex cube root of unity, the value of $\frac{(\omega^2 - x_1)(\omega^2 - x_2)(\omega^2 - x_3)}{(\omega - x_1)(\omega - x_2)(\omega - x_3)}$ is

29. The number of common roots of the equations $x^5 - x^3 + x^2 - 1 = 0$ and $x^4 - 1 = 0$ is

30. If $z_n = (1 + i\sqrt{3})^n$ and the value of $\sqrt{3} \operatorname{Im}(z_5 \bar{z}_4) = \lambda$ then the integral part of $\frac{\lambda}{100}$ is

31. If Z_1 and $\bar{Z}_1$ represents adjacent vertices of a regular polygon of n sides and if $\frac{\operatorname{Im}(Z_1)}{\operatorname{Re}(Z_1)} = \sqrt{2} - 1$ then n is equal to

24. If z_1, z_2, z_3, z_4 are the roots of the equation $z^4 + z^3 + z^2 + z + 1 = 0$, then

COLUMN - I

COLUMN - II

A) $\left| \sum_{i=1}^4 z_i^4 \right|$ is equal to

p) 0

B) $\sum_{i=1}^4 z_i^5$ is equal to

q) 4

C) $\prod_{i=1}^4 (z_i + 2)$ is equal to

r) 1

D) Least value of $[z_1 + z_2]$ is

s) 11

where $[]$ represents greatest integer function

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A) Let $z_k (k=0,1,2,3,4,5,6)$ be the roots of the equation

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$(z+1)^7 + (z)^7 = 0$ then $\sum_{k=0}^6 \operatorname{Re}(z_k)$

B) If α, β, γ and a, b, c are complex numbers such that

q) $-\frac{7}{2}$

$\frac{\alpha}{a}, \frac{\beta}{b}, \frac{\gamma}{c} = 1+i$ and $\frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma} = 0$, then the value of $\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2} + \frac{\gamma^2}{c^2}$

C) If $z_1, z_2, \dots, z_6$ are six roots of the equations

r) 1

$z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = 0$, then the value of $\prod_{i=1}^6 (z_i + 1)$

D) If $|z| = \min(|z-1|, |z+1|)$, then the value of $|z + \bar{z}|$

s) 4

Integer answer type questions

26. If $a_1, a_2, \dots, a_n$ are real numbers and $\cos \alpha + i \sin \alpha$ is a root of $x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n = 0$ then the value of $|a_1 \cos \alpha + a_2 \cos 2\alpha + \dots + a_n \cos(n\alpha)|$ is

27. If $z = \frac{1}{2}(\sqrt{3} - i)$ and the smallest value of the positive integer n for which $(z^{89} + i^{97})^{94} = z^n$ is λ then $\frac{\lambda}{2} =$

28. If $1, x_1, x_2, x_3$ are the roots of $x^4 - 1 = 0$ and ω is a complex cube root of unity, the value of $\frac{(\omega^2 - x_1)(\omega^2 - x_2)(\omega^2 - x_3)}{(\omega - x_1)(\omega - x_2)(\omega - x_3)}$ is

29. The number of common roots of the equations $x^5 - x^3 + x^2 - 1 = 0$ and $x^4 - 1 = 0$ is

30. If $z_n = (1 + i\sqrt{3})^n$ and the value of $\sqrt{3} \operatorname{Im}(z_5 \bar{z}_4) = \lambda$ then the integral part of $\frac{\lambda}{100}$ is

31. If Z_1 and $\bar{Z}_1$ represents adjacent vertices of a regular polygon of n sides and if $\frac{\operatorname{Im}(Z_1)}{\operatorname{Re}(Z_1)} = \sqrt{2} - 1$ then n is equal to

EXERCISE-III

Geometrical Applications

LEVEL-I (MAIN)

Single answer type questions

- Let A, B and C represents the complex numbers z_1, z_2 and z_3 in the Argand plane. If circumcentre of the triangle ABC is at the origin, then the complex number corresponding to orthocentre is
 - $\frac{1}{4}(z_1 + z_2 + z_3)$
 - $\frac{1}{3}(z_1 + z_2 + z_3)$
 - $\frac{1}{2}(z_1 + z_2 + z_3)$
 - $z_1 + z_2 + z_3$
- If $z = x + iy$ then the equation $\left| \frac{2z-i}{z+1} \right| = m$ does not represent a circle when
 - $m = \frac{1}{2}$
 - $m = 1$
 - $m = 2$
 - $m = 3$
- If $|z| = \sqrt{2}$ then the point given by " $3+4z$ " lies on a circle whose radius is
 - $2\sqrt{2}$
 - $3\sqrt{2}$
 - $4\sqrt{2}$
 - $5\sqrt{2}$
- $z = x+iy$ and $w = \frac{1-iz}{z-i}$, then $|w| = 1 \Rightarrow$ in the complex plane
 - z lies on the imaginary axis
 - z lies on the real axis
 - z lies on the unit circle
 - z lies on the parabola
- If $|z + \bar{z}| + |z - \bar{z}| = 2$ then z lies on
 - a straight line
 - a square
 - a circle
 - parallelogram
- If $z(\bar{z} + 3) = 2$ then the locus of $z = x+iy$ is
 - $x^2 + y^2 + 3x - 2 = 0, y = 0$
 - $x = 0$ such that $y > 2/3$
 - $x^2 + y^2 + 2x - 4y = 0$ such that $y < 0, x^2 + y^2 > 1$
 - $x^2 + y^2 + 2x - 4y = 0$ such that $2x - y + 4 > 0$
- The locus of z satisfying the inequality $\log_{\frac{1}{3}} |z+1| > \log_{\frac{1}{3}} |z-1|$ is
 - $\operatorname{Re}(z) > 0$
 - $\operatorname{Re}(z) < 0$
 - $\operatorname{Im}(z) > 0$
 - $\operatorname{Im}(z) < 0$
- If $z_1 = 8+4i, z_2 = 6+4i$ and z be a complex number such that $\operatorname{Arg} \left(\frac{z-z_1}{z-z_2} \right) = \frac{\pi}{4}$, then the locus of z is
 - $(x-7)^2 + (y-5)^2 = 2$
 - $(x-7)^2 + (y+5)^2 = 2$
 - $(x-7)^2 + (y-5)^2 = 4$
 - $(x+7)^2 + (y+5)^2 = 4$

9. If $|z| = 3$, the area of the triangle whose sides are z , $z\omega$ and $z+\omega z$ (where ω is a complex cube root of unity) is
- 1) $\frac{9\sqrt{3}}{4}$ 2) $\frac{8\sqrt{3}}{2}$ 3) $\frac{5}{2}$ 4) $\frac{8\sqrt{3}}{3}$
10. Let z_1 and z_2 be two non-zero complex numbers such that $\frac{z_1}{z_2} + \frac{z_2}{z_1} = 1$, then the origin and points represented by z_1 and z_2
- 1) Lie on a straight line 2) Form a right triangle
3) Form an equilateral triangle 4) form an isosceles triangle
11. If a and b are real numbers between 0 and 1 such that the points $z_1 = a+bi$, $z_2 = 1+bi$ and $z_3 = 0$ form an equilateral triangle then a, b are
- 1) $2-\sqrt{3}, 2-\sqrt{3}$ 2) $2-\sqrt{3}, 2+\sqrt{3}$ 3) $2+\sqrt{3}, 2-\sqrt{3}$ 4) $2-\sqrt{2}, 2+\sqrt{2}$
12. The roots of the cubic equation $(z+\alpha\beta)^3 = \alpha^3, \alpha \neq 0$ represent the vertices of a triangle of sides of length
- 1) $\frac{1}{\sqrt{3}}|\alpha\beta|$ 2) $\sqrt{3}|\alpha|$ 3) $\sqrt{3}|\beta|$ 4) $\frac{1}{\sqrt{3}}|\alpha|$
13. The equation $\|z+i\| - \|z-i\| = k$ represents a hyperbola if
- 1) $-3 < k < 2$ 2) $k > 2$ 3) $0 < k < 2$ 4) $k > 4$
14. Let $z = 1-t+i\sqrt{t^2+t+2}$, where t is a real parameter. The locus of z in the Argand plane is
- 1) a hyperbola 2) an ellipse 3) a straight line 4) a circle

Numerical value type questions

15. One vertex of an equilateral triangle is at the origin and the other two vertices are, roots of $2z^2 + 2z + k = 0$, then k is
16. Let $Z = 1-t+i\sqrt{t^2+t+2}$ where t is a real parameter. Locus of Z is a conic then eccentricity of conic is
17. If $az^2+bz+1=0$ where $a, b, c \in \mathbb{Z}$ $|a| = \frac{1}{2}$ and have a root α such that $|\alpha| = 1$ and $|\overline{ab} - b| =$
18. Let z and ω be complex numbers, If $\operatorname{Re}(z) = |z-2|$, $\operatorname{Re}(\omega) = |\omega-2|$ and $\operatorname{Arg}(z-\omega) = \frac{\pi}{3}$, the value of $\frac{\operatorname{Im}(z+\omega)}{\sqrt{3}}$ is

LEVEL-II (ADVANCED)**Single answer type questions**

1. If z_1, z_2, z_3 and z_4 be the consecutive vertices of a square, then $z_1^2 + z_2^2 + z_3^2 + z_4^2$ equals
- a) $z_1 z_2 + z_2 z_3 + z_3 z_4 + z_4 z_1$ b) $z_1 z_2 + z_1 z_3 + z_1 z_4 + z_2 z_3 + z_2 z_4 + z_3 z_4$
c) 0 d) None of the above

2. If z_1, z_2 and z_3 are the vertices of an isosceles right angled triangle, right angled at the vertex z_2 , then $(z_1 - z_2)^2 + (z_3 - z_2)^2$ equals
 - a) 0
 - b) $(z_1 - z_3)^2$
 - c) $\left(\frac{z_1 + z_3}{2}\right)^2$
 - d) None of these
3. The point of intersection of the curves $\arg(z - 3i) = \frac{3\pi}{4}$ and $\arg(2z + 1 - 2i) = \frac{\pi}{4}$ is
 - a) $\frac{1}{4}(3 + 9i)$
 - b) $\frac{1}{4}(3 - 9i)$
 - c) $\frac{1}{2}(3 + 2i)$
 - d) None of these.
4. If z_1, z_2, z_3 are non-zero complex numbers representing the points A, B, C such that $\frac{2}{z_1} = \frac{1}{z_2} + \frac{1}{z_3}$. Then
 - a) A, B, C are collinear.
 - b) Circle passes through points A, B, C has centre at origin O
 - c) Circle passes through A, B, C passes through origin.
 - d) None of these.
5. If $|2z - 4 - 2i| = |z| \sin\left(\frac{\pi}{4} - \arg z\right)$, then locus of z is/an
 - a) Ellipse
 - b) Circle
 - c) Parabola
 - d) Pair of straight line
6. If $|z - 3i| = 3$, (where $i = \sqrt{-1}$) and $\arg z \in (0, \pi/2)$, then $\cot(\arg(z)) - \frac{6}{z}$ is equal to
 - a) 0
 - b) $-i$
 - c) i
 - d) none of these
7. The complex number $3 + 4i$ is rotated about origin by an angle of $\frac{\pi}{4}$ and then stretched 2 times. The complex number corresponding to new position is
 - a) $\sqrt{2}(-3 + 4i)$
 - b) $\sqrt{2}(-1 + 7i)$
 - c) $\sqrt{2}(3 - 4i)$
 - d) $\sqrt{2}(-1 - 7i)$
8. If the complex numbers z_1, z_2, z_3 satisfying $3z_1 = 5z_2 - 2z_3$, then z_1, z_2 and z_3 lie in a
 - a) circle
 - b) parabola
 - c) line
 - d) hyperbola
9. The value of $\sin\left[\log_e\left\{\left(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2}\right)^z\right\}\right]$ is, where z satisfies the equation $|z - 2i| = 1$ and has least modulus
 - a) 1
 - b) 0
 - c) -1
 - d) $\frac{1}{2}$
10. Let a, b, c be three points lying on the circle $|z| = 1$ and suppose $\alpha \in (0, \pi/2)$ be such that $a + b \cos \alpha + c \sin \alpha = 0$, then
 - a) $b = \pm ic$
 - b) $2a^2 + b^2 + c^2 = 0$
 - c) $a^2 + b^2 + c^2 = 0$
 - d) none of these
11. Let $z_1, z_2, z_3 \in \mathbb{C}$ such that $z_1 + z_2 + z_3 = 0$ and $|z_1| = |z_2| = |z_3| = 1$ then $|z_1^2 + 2z_2^2 + z_3^2| =$
 - a) 1
 - b) 2
 - c) 3
 - d) 4

12. If $w \neq 1$ is a cube root of 1 and $|z-1|^2 + 2|z-w|^2 = 3|z-w^2|^2$ then z lies on
- a straight line
 - a parabola
 - An ellipse
 - a rectangular hyperbola
13. If $z \neq 1$, and $\frac{z^2}{z-1}$ is real, then the point represented by the complex number z lies
- on a circle with centre at origin
 - either on real axis or on a circle not passing through the origin
 - on imaginary axis
 - either on the real axis or on a circle passing through origin
14. Locus of z if $\arg(z-(1+i)) = \begin{cases} \frac{3\pi}{4} & \text{if } |z| \leq |z-2| \\ -\frac{\pi}{4} & \text{if } |z| > |z-2| \end{cases}$ is
- straight lines passing through (2,0)
 - straight lines passing through (2,0), (1,1)
 - a line segment
 - a set of two rays
15. Let S_1 and S_2 are concentric circles with radius 1 and $8/3$ respectively, having centre at (3,0), on the argand plane. If z satisfies the inequality $\log_3 \left(\frac{|z-3|^2 + 2}{11|z-3|-2} \right) > 1$ then
- z lies outside S_1 but inside S_2
 - z lies inside of both S_1 and S_2
 - z lies outside both of S_1 and S_2
 - z lies on S_1
16. $P = \{z : \operatorname{Im} z \geq 1\}$; $Q = \{z : |z-2-i| = 3\}$; $R = \{z : \operatorname{Re}(1-i)z = \sqrt{2}\}$. If $Z \in P \cap Q \cap R$ then $|z+1-i|^2 + |z-5-i|^2 =$
- 23
 - 36
 - 72
 - 34
17. If $z_1, z_2, z_3, \dots, z_{10}$ denote vertices of regular polygon of 10 sides with z_0 as centre of polygon then $\frac{z_1^2 + z_2^2 + z_3^2 + \dots + z_{10}^2}{z_0^2}$ equals to
- 1
 - 10
 - 100
 - 2
18. The length of the curve which is the locus represented by $\arg\left(\frac{z+i}{z-i}\right) = \frac{\pi}{4}$ is equal to
- $\frac{3\pi}{2}$
 - $\frac{3\pi}{\sqrt{2}}$
 - $\frac{\pi}{\sqrt{2}}$
 - $\pi\sqrt{2}$
19. The image of complex number z satisfying $\arg\left(\frac{z-i}{z-3}\right) = \frac{\pi}{6}$, about real axis is
- $\arg\left(\frac{z+i}{z-3}\right) = \frac{\pi}{6}$
 - $\arg\left(\frac{z-i}{z+3}\right) = \frac{\pi}{6}$
 - $\arg\left(\frac{z+i}{z-3}\right) = \frac{-\pi}{6}$
 - $\arg\left(\frac{z-i}{z+3}\right) = \frac{-\pi}{6}$
20. If ω is the complex cube root of unity, then the $\arg(i\omega) + \arg(i\omega^2)$ is
- 0
 - $\frac{\pi}{2}$
 - π
 - $\frac{\pi}{3}$

21. The locus of z such that $\arg\left(\frac{z-z_1}{z-z_2}\right) = \frac{\pi}{4}$ where $z_1 = 10+6i, z_2 = 4+6i$ is part of a circle whose radius is

a) $\sqrt{2}$ b) $2\sqrt{2}$ c) $3\sqrt{2}$ d) 1

More than one correct answer type questions

22. Let z_1, z_2, z_3 be the complex numbers representing the vertices A, B, C of a triangle described in counterclock sense respectively. Consider the following statements.

I) $\triangle ABC$ is equilateral II) $z_3 - z_1 = (z_2 - z_1)\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$

III) $z_2 - z_1 = (z_3 - z_1)\left(\cos\frac{5\pi}{3} + i\sin\frac{5\pi}{3}\right)$

IV) $z_1 + z_2\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right) + z_3\left(\cos\frac{4\pi}{3} + i\sin\frac{4\pi}{3}\right) = 0$

Then which one is correct:

a) $I \Rightarrow II$ b) $II \Rightarrow III$ c) $III \Rightarrow IV$ d) $IV \Rightarrow I$

23. If from a point P representing the complex numbers z_1 on the curve $|z|=2$, pair of tangents are drawn to the curve $|z|=1$, meeting at points $Q(z_2)$ and $R(z_3)$, then

a) complex number $\frac{z_1 + z_2 + z_3}{3}$ will lie on the curve $|z|=1$

b) $\left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right)\left(\frac{4}{z_1} + \frac{1}{z_2} + \frac{1}{z_3}\right) = 9$ c) $\arg\left(\frac{z_2}{z_3}\right) = \pm\frac{2\pi}{3}$

d) orthocentre and circumcentre of $\triangle PQR$ will coincide

24. The equations of two lines making an angle 45° with a given line $\bar{a}z + a\bar{z} + b = 0$ (where ' a ' is a complex number and b is real) and passing through a given point C (c is a complex number), is/are

a) $\frac{z+c}{a} + i\frac{\bar{z}-\bar{c}}{\bar{a}} = 0$

b) $\frac{z-c}{a} + i\frac{\bar{z}-\bar{c}}{\bar{a}} = 0$

c) $\frac{z-c}{a} - i\frac{\bar{z}-\bar{c}}{\bar{a}} = 0$

d) $\frac{z+c}{a} - i\frac{\bar{z}-\bar{c}}{\bar{a}} = 0$

25. Let z_1 and z_2 are non-zero (given) complex numbers and k be any positive real number. Consider the system of equations $|3z - z_1 - 2z_2| = |z_1 - z_2|$ and $\arg\left(\frac{z_1 - z_2}{z - kz_1 - (1-k)z_2}\right) = \pm\frac{\pi}{2}$. Then

a) The system of equations has no solution if $k \in \left(\frac{2}{3}, \infty\right)$

b) The system of equations have more than one solutions if $k \in \left(0, \frac{2}{3}\right)$

c) The system of equations have no solution if $k \in \left(0, \frac{2}{3}\right)$

d) The system of equations have more than one solution if $k \in \left(\frac{2}{3}, \infty\right)$

26. If α is a variable complex number such that $|\alpha| > 1$ ($|\alpha|$ is a constant) and $z = \alpha + \frac{1}{\alpha}$ lies on a conic then

- a) Eccentricity of the conic is $\frac{2|\alpha|}{1+|\alpha|^2}$ b) Distance between foci is 4
 c) Length of latusrectum is $\frac{2(|\alpha|^2-1)}{|\alpha|^2+1}$ d) Distance between directrices is $\left(|\alpha| + \frac{1}{|\alpha|}\right)^2$

27. Let a, b, c be distinct complex numbers with $|a| = |b| = |c| = 1$ and z_1, z_2 be the roots of the equation $az^2 + bz + c = 0$ with $|z_1| = 1$. Let P and Q represent the complex numbers z_1 and z_2 in the argand plane with $\angle POQ = \theta$, $0^\circ < \theta < 180^\circ$ (where O being the origin) then

- a) $b^2 = ac$; $\theta = \frac{2\pi}{3}$ b) $\theta = \frac{2\pi}{3}$; $PQ = \sqrt{3}$ c) $PQ = 2\sqrt{3}$; $b^2 = ac$ d) $\theta = \frac{\pi}{3}$; $b^2 = ac$

Linked comprehension type questions

Passage-I :

In acute angle $\triangle ABC$ altitude are drawn from vertices A, B, C which meets sides BC, AC, AB at $D(z_1), E(z_2), F(z_3)$ such that $\arg\left(\frac{z_3 - z_2}{z_2 - z_1}\right)$ purely img. If $BC = 4$ units

28. Circum radius of $\triangle ABC$ is

- a) $3\sqrt{2}$ b) $2\sqrt{2}$ c) 2 d) None of these

29. The length of EF

- a) $2\sqrt{2}$ b) $\sqrt{2}$ c) 2 d) None of these

30. Circum radius of $\triangle DEF$

- a) 4 b) 2 c) $2\sqrt{2}$ d) None

Passage-II :

Tangents are drawn to a point P to circle $|z - (1+i)| = 4$; such that angle between them is $\frac{\pi}{3}$, intersect the circle at A and B . If I is incentre of $\triangle PAB$.

31. Area of $\triangle IAB$

- a) $3\sqrt{3}$ b) $4\sqrt{3}$ c) 4 d) None

32. Circum radius of $\triangle IAB$

- a) 2 b) 4 c) $2\sqrt{2}$ d) None

33. Area of $\triangle PAB$

- a) $12\sqrt{3}$ b) $4\sqrt{3}$ c) $8\sqrt{3}$ d) None

Passage-III :

If Z_1, Z_2 be the complex numbers representing two points A and B then we define complex slope of line AB as $\mu = \frac{Z_1 - Z_2}{\bar{Z}_1 - \bar{Z}_2}$. It can be noted that $|\mu| = 1$ and μ remains same for any two points on the line AB . Now answer the following questions.

34. The complex slope of line $a\bar{Z} + \bar{a}Z + b = 0$ where a is complex and b is real is
 a) $\frac{a}{\bar{a}}$ b) $-\frac{a}{\bar{a}}$ c) $\frac{\bar{a}}{a}$ d) $-\frac{\bar{a}}{a}$
35. If a line in Argand plane passes through two points which are represented by Z_1, Z_2 and the line has complex slope μ then the equation of line must be of the form
 a) $Z - Z_1 = \mu(Z - Z_2)$ b) $Z - Z_1 = \mu(\bar{Z} - \bar{Z}_1)$ c) $Z - Z_1 = \frac{1}{\mu}(\bar{Z} - \bar{Z}_1)$ d) $\frac{Z - Z_1}{Z - Z_2} = \mu + \frac{Z - Z_2}{Z - Z_1}$
36. If the complex slope of a line passing through origin which is not parallel to y -axis is $\cos\phi + i\sin\phi$ then the line makes angle θ with x -axis such that
 a) $\theta = 2\phi$ b) $\theta = 90^\circ - \phi$ c) $\theta = \phi/2$ d) $\theta = \phi$

Passage-IV :

A person walks $2\sqrt{2}$ units away from origin in south west direction to reach A , then walks $\sqrt{2}$ units in south east ($S45^\circ E$) direction to reach B . From B he travels 4 units horizontally towards east to reach C , then he travels along a circular path with centre at origin through an angle of $\frac{2\pi}{3}$ in anti clockwise direction to reach his destination D .

37. Position of B in argand plane is
 a) $\sqrt{2}e^{-i\frac{3\pi}{4}}$ b) $\sqrt{2}(2+i)e^{-i\frac{3\pi}{4}}$ c) $\sqrt{2}(1+2i)e^{-i\frac{3\pi}{4}}$ d) $-3+i$
38. Let the complex number Z represents C in argand plane then $\arg(Z) =$
 a) $-\frac{\pi}{6}$ b) $\frac{\pi}{4}$ c) $-\frac{\pi}{4}$ d) $\frac{\pi}{3}$

Matrix matching type questions

39. COLUMN - I

COLUMN - II

- | | |
|---|----------------------------|
| A) Locus of the point ' z ' satisfying the equation $\operatorname{Re}(z^2) = \operatorname{Re}(z + \bar{z})$ | p) A parabola |
| B) Locus of the point ' z ' satisfying the $ z - z_1 + z - z_2 = \lambda, \lambda \in \mathbb{R}^+$ and $\lambda z_1 - z_2 $ | q) A straight line |
| C) Locus of the point ' z ' satisfying the $\left \frac{2z-i}{z+1} \right = m$, Where $i = \sqrt{-1}$ and $m \in \mathbb{R}^+$ | r) An ellipse |
| D) If $ z = 25$, then the points representing the complex number $-1 + 75\bar{z}$ will be on ' a ' | s) A rectangular hyperbola |
| | t) A circle |

40. Which of the following condition/s in column-II are satisfied by the quadrilateral formed by Z_1, Z_2, Z_3, Z_4 in order given in column-I.

COLUMN - I

- A) Parallelogram
B) Rectangle
C) Rhombus
D) Square

COLUMN - II

- p) $Z_1 - Z_4 = Z_2 - Z_3$
q) $|Z_1 - Z_3| = |Z_2 - Z_4|$
r) $\frac{Z_1 - Z_2}{Z_3 - Z_4}$ is purely real
s) $\frac{Z_1 - Z_3}{Z_2 - Z_4}$ is purely imaginary
t) $\frac{Z_1 - Z_2}{Z_3 - Z_4}$ is purely imaginary

Integer answer type questions

41. If the points $1 + 2i$ and $-1 + 4i$ are reflections of each other in the line $z(1+i) + \bar{z}(1-i) + K = 0$, then the value of K is _____.
42. If the area of a triangle with vertices Z_1, Z_2 and Z_3 is the absolute value of the number $\lambda i \begin{vmatrix} Z_1 & \bar{Z}_1 & 1 \\ Z_2 & \bar{Z}_2 & 1 \\ Z_3 & \bar{Z}_3 & 1 \end{vmatrix}$, then the value of $1/\lambda$ is equal to _____.
43. If the complex number z is such that $|z-1| \leq 1$ and $|z-2|=1$, then the maximum possible value $|z|^2$ is _____.
44. If $0 \leq \arg(z) \leq \pi/4$, then the least value of $\sqrt{2}|z-i|$ is _____.
45. The number of points z in the complex plane satisfying both the equations $|z-4-8i| = \sqrt{10}$ and $|z-3-5i| + |z-5-11i| = 4\sqrt{5}$ is _____.
46. Let $\frac{1}{a_1-2i}, \frac{1}{a_2-2i}, \frac{1}{a_3-2i}, \frac{1}{a_4-2i}, \frac{1}{a_5-2i}, \frac{1}{a_6-2i}, \frac{1}{a_7-2i}, \frac{1}{a_8-2i}$ are vertices of regular octagon. If the area of octagon is A , then the value of $8\sqrt{2}A$ is (where $a_j \in \mathbb{R}$ for $j = 1, 2, 3, 4, 5, 6, 7, 8$ and $i = \sqrt{-1}$) _____.
47. Let $1, w, w^2$ be the cube roots of unity. The least possible degree of a polynomial with real coefficients having roots $2w, (2+3w), (2+3w^2), (2-w-w^2)$, is _____.

KEY SHEET (LECTURE SHEET)

EXERCISE-I

LEVEL-I

- 1) 2 2) 2 3) 2 4) 1 5) 4 6) 1 7) 1 8) 3
 9) 1 10) 4 11) 3 12) 3 13) 1 14) 2 15) 1 16) 6
 17) 2 18) 4 19) 3364 20) 1.4 21) 2022 22) 9 23) 0.2

LEVEL-II

- 1) b 2) a 3) c 4) a 5) c 6) d 7) b 8) b
 9) d 10) a 11) b 12) c 13) a 14) a 15) b 16) c
 17) b 18) b 19) a 20) b 21) a 22) b 23) ad 24) abcd
 25) ab 26) ad 27) abc 28) abd 29) bc 30) abcd 31) abc 32) c
 33) a 34) b 35) c 36) a 37) a 38) b 39) d 40) d
 41) c 42) b 43) a 44) A-r;B-p;C-s;D-q 45) A-r;B-p;C-p;D-q
 46) 3 47) 8 48) 6 49) 4 50) 2 51) 2

EXERCISE-II

LEVEL-I

- 1) 1 2) 2 3) 1 4) 1 5) 2 6) 4 7) 1 8) 2
 9) 2 10) 2 11) 3 12) 2 13) 1 14) 1 15) 2 16) 4
 17) 2 18) 1 19) 1 20) 2 21) 1 22) 3 23) 2 24) 2
 25) 1 26) 2 27) 3 28) 2 29) 3 30) 0 31) 0.04 32) 0.5
 33) 1 34) 0.25

LEVEL-II

- 1) a 2) c 3) b 4) c 5) b 6) c 7) d 8) b
 9) d 10) d 11) d 12) abc 13) cd 14) ab 15) abcd
 16) c 17) c 18) b 19) d 20) c 21) c 22) c
 23) c 24) A-r;B-qt;C-s;D-p 25) A-q;B-p;C-s;D-r 26) 1
 27) 5 28) 1 29) 2 30) 7 31) 8

EXERCISE-III

LEVEL-I

- 1) 4 2) 3 3) 3 4) 2 5) 2 6) 1 7) 2 8) 1
 9) 1 10) 3 11) 1 12) 2 13) 3 14) 1 15) 0.67 16) 1.41
 17) 0.75 18) 1.33

LEVEL-II

- 1) a 2) a 3) d 4) c 5) a 6) c 7) b 8) c
 9) c 10) a 11) a 12) a 13) d 14) d 15) a 16) b
 17) b 18) d 19) c 20) c 21) c 22) abcd 23) abcd 24) bc
 25) ab 26) abd 27) ab 28) c 29) a 30) c 31) a 32) b
 33) a 34) b 35) b 36) c 37) c 38) c
 39) A-p;B-q;C-r;D-st 40) A-pr;B-pqrt;C-prs;D-pqrst
 41) 6 42) 4 43) 3 44) 1 45) 2 46) 2 47) 5

COMPLEX NUMBERS

LECTURE SHEET

EXERCISE-I

LEVEL-I (MAIN)

Single answer type questions

$$1. \quad x = \frac{3(2 + \cos \theta)}{5 + 4 \cos \theta} \Rightarrow (x-1)(x-3) = \frac{-9 \sin^2 \theta}{5 + 4 \cos \theta} = -y^2$$

$$2. \quad x + iy = a^3 - 3ab^2 + i(3a^2b - b^3) \\ \Rightarrow \frac{x}{a} = a^2 - 3b^2; \frac{y}{b} = 3a^2 - b^2$$

$$3. \quad \text{Imaginary part} = 0$$

$$\tan \frac{x}{2} - 2 \sin \left(\frac{x}{2} \right) \left[\sin \frac{x}{2} + \cos \frac{x}{2} \right] = 0$$

$$4. \quad \text{Take modulus on both sides}$$

$$5. \quad \text{Take modulus on both sides}$$

$$6. \quad Z \text{ is unimodular} \Rightarrow \bar{z} = \frac{1}{z} \text{ and } |z| = |\bar{z}|$$

$$7. \quad |z_1 - 3z_2| = |3 - z_1 \bar{z}_2| \text{ use } z \bar{z} = |z|^2$$

$$8. \quad \sqrt{(a+b)^2 - (a-b)^2 - 2(a+b)(a-b)i} \\ = \sqrt{[(a+b) - i(a-b)]^2} = \pm[(a+b) - i(a-b)]$$

$$9. \quad x^4 + x^2 + 1 = (x^2 + 1)^2 - x^2 = (x^2 + x + 1)(x^2 - x + 1)$$

$$10. \quad x + iy = a^2 - b^2 + i2ab \Rightarrow x = a^2 - b^2; \\ y = 2ab - x - iy = b^2 - a^2 - i2ab = (b - ia)^2$$

$$11. \quad \left| \frac{z_1 + z_2}{z_1 - z_2} \right| = 1; \left| \frac{\frac{z_1}{z_2} + 1}{\frac{z_1}{z_2} - 1} \right| = 1; \left| \frac{\frac{z_1}{z_2} + 1}{\frac{z_1}{z_2} - 1} \right| = \left| \frac{\frac{z_1}{z_2} - 1}{\frac{z_1}{z_2} + 1} \right| \\ \Rightarrow \frac{z_1}{z_2} = 0 \text{ or purely imaginary}$$

$$12. \quad \text{We have } |z_1 + z_2 + z_3 - 6 + 6i| \\ = |(z_1 - 1) + (z_2 - 2) + (z_3 - 3) + 6i| \\ \leq |z_1 - 1| + |z_2 - 2| + |z_3 - 3| + 6 < 1 + 2 + 3 + 6 = 12.$$

$$13. \quad z = re^{i\theta}, \bar{z} = re^{-i\theta}; \left| \frac{z}{r} - \frac{\bar{z}}{r} \right| = |2i \sin \theta| \leq |\arg z - \arg \bar{z}| \\ \Rightarrow |z - \bar{z}| \leq |z| |\arg z - \arg \bar{z}|$$

$$14. \quad \bar{z} = \begin{vmatrix} 1 & 1+2i & 3-5i \\ 1-2i & -5 & -10i \\ 3+5i & -10i & 11 \end{vmatrix} = z \quad \therefore z \text{ is purely real.}$$

$$15. \quad i = e^{i\pi/2} \Rightarrow i^i = e^{-\pi/2} = k \text{ (say) then}$$

$$i^i = i^k \Rightarrow |i^i| = \left| e^{-\frac{\pi}{2}} \right| = 1$$

Numerical value type questions

$$16. \quad \text{min occurs at median of } 1, 2, 3, 4, 5 \text{ i.e. at '3'}$$

$$17. \quad e^{iz} = e^{-\log(2-\sqrt{3})} = 2 + \sqrt{3}$$

$$\cos z = \frac{e^{iz} + e^{-iz}}{2} = 2$$

$$18. \quad OA = |4+2i-iz| = |4| \quad |4i-2+z|$$

$$|z+i-2+3i| \leq |z+i-2| + |3i| \leq 4$$

$$19. \quad z = (3\lambda - 7\mu) + i(7\lambda + 3\mu)$$

$$\therefore 3\lambda - 7\mu = 0$$

$$\frac{\lambda}{\mu} = \frac{7}{3}$$

$$\text{From minimum, } \lambda = 7, \mu = 3$$

$$|z|^2 = |3+7i|^2 = |3+7i|^2 = 58(\lambda^2 + \mu^2)$$

$$20. \quad \text{If } \left| z - \frac{a}{z} \right| = b \text{ then}$$

$$\frac{-b + \sqrt{b^2 + 4a}}{2} \leq |z| \leq \frac{b + \sqrt{b^2 + 4a}}{2}$$

$$\therefore \lambda = 6, \mu = 1$$

$$\frac{\lambda^\mu + \mu^\lambda}{\lambda^\mu - \mu^\lambda} = \frac{7}{5} = 1.4$$

$$21. \quad \text{Let } z = a + ib, \bar{z} = a - ib$$

$$Z^{2020} = \bar{z}$$

$$|z|^{2020} = |\bar{z}| = |z|$$

$$\Rightarrow |z| = 0 \text{ or } |z|^{2019} = 1$$

$$|z| = 0 \Rightarrow z = 0$$

$$\text{If } |z| = |z|^{2021} = z \bar{z} = |z|^2 = 1$$

$$2021 \text{ values of } z$$

$$\text{Total} = 2022$$

22. Let $y = |z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_1|^2$
 $= (z_1 - z_2)(\bar{z}_1 - \bar{z}_2) + (z_2 - z_3)(\bar{z}_2 - \bar{z}_3) + (z_3 - z_1)(\bar{z}_3 - \bar{z}_1)$
 $= 6 - (z_1\bar{z}_2 + z_2\bar{z}_1 + \bar{z}_2z_3 + z_3\bar{z}_2 + z_1\bar{z}_3 + z_3\bar{z}_1) \quad (1)$
 we know that $|z_1 + z_2 + z_3|^2 \geq 0$
 $\Rightarrow 3 + (z_1\bar{z}_2 + z_2\bar{z}_1 + z_1\bar{z}_3 + z_3\bar{z}_1 + z_2\bar{z}_3 + z_3\bar{z}_2) \geq 0 \quad (2)$
 from (1) and (2) $y \leq 9$

23. $\frac{a}{a_1} + \frac{b}{b_1} + \frac{c}{c_1} = 1 + i$
 $\Rightarrow \frac{a^2}{a_1^2} + \frac{b^2}{b_1^2} + \frac{c^2}{c_1^2} + 2\left(\frac{ab}{a_1b_1} + \frac{bc}{b_1c_1} + \frac{ca}{c_1a_1}\right) = 2i$
 $\Rightarrow \frac{a^2}{a_1^2} + \frac{b^2}{b_1^2} + \frac{c^2}{c_1^2} + 2\frac{abc}{a_1b_1c_1}\left(\frac{c_1}{c} + \frac{a_1}{a} + \frac{b_1}{b}\right) = 2i$
 $\frac{\left|\frac{a^2}{a_1^2} + \frac{b^2}{b_1^2} + \frac{c^2}{c_1^2}\right|}{1000} = \frac{2}{1000} = 0.002$

LEVEL-II (ADVANCED)

Single answer type questions

- $\Rightarrow a + ib + \sqrt{a^2 + b^2} = 2 + 8i$
 $\Rightarrow b = 8, a + \sqrt{a^2 + 64} = 2$
 $a^2 + 64 = a^2 - 4a + 4 \Rightarrow a = -15$
- $|3Z + 9 - 7i| = |3Z + 6 - 3i + 3 - 4i|$
 $\leq 3|z + 2 - i| + \sqrt{3^2 + 4^2} \Rightarrow 3(5) + 5 = 20$
- $\bar{z} + i\bar{w} = 0 \Rightarrow z = iw \quad \arg(z) + \arg(w) = \pi$
 $\Rightarrow \arg(iw) + \arg w = \pi$
 $\Rightarrow \frac{\pi}{2} + 2\arg w = \pi \Rightarrow \arg w = \frac{\pi}{4} \Rightarrow \arg(z) = \frac{3\pi}{4}$
- squaring and adding both given eq
 $(z_1^2 + z_2^2)^3 = 125 \quad z_1^2 + z_2^2 = 5$
- We have
 $\left|\frac{z_1}{2} + \frac{z_2}{3} + \frac{z_3}{4} + \frac{z_4}{5}\right| = \frac{k}{60} |z_1z_2z_3z_4| \left|\frac{1}{z_1} + \frac{1}{z_2} + \frac{1}{z_3} + \frac{1}{z_4}\right|$
 Now, $z_1\bar{z}_1 = 2, z_2\bar{z}_2 = 3, z_3\bar{z}_3 = 4$ and $z_4\bar{z}_4 = 5$
 So, $k = \frac{60}{|z_1z_2z_3z_4|} = \frac{60}{\sqrt{2}\sqrt{3}\sqrt{4}\sqrt{5}} = \sqrt{30} = |z_4z_1z_2|$

- $|8z_2z_3 + 27z_3z_1 + 64z_1z_2| = |z_1||z_2||z_3| \left|\frac{8}{z_1} + \frac{27}{z_2} + \frac{64}{z_3}\right|$
 $= (2)(3)(4) \left|\frac{8\bar{z}_1}{|z_1|^2} + \frac{27\bar{z}_2}{|z_2|^2} + \frac{64\bar{z}_3}{|z_3|^2}\right|$
 $= (24)(4) = 96$
- $0 = \frac{1-z}{1+z} + \frac{1-\bar{z}}{1+\bar{z}} = \frac{2(1-|z|^2)}{1+|z|^2} \Rightarrow |z| = 1$
- $|z_1 + z_2| = |z_1 - z_2| \Rightarrow \frac{z_1}{z_2}$ is purely imaginary $\arg \frac{z_1}{z_2} = \frac{\pi}{2}$
 $\Rightarrow$ Circumcentre of POQ is the mid point of PQ
- $\frac{z_1 + z_2}{z_1 - z_2} = \cos \alpha + i \sin \alpha \quad \frac{z_1}{z_2} = \frac{1 + \cos \alpha + i \sin \alpha}{-1 + \cos \alpha + i \sin \alpha}$
 $= -i \cot \left(\frac{\alpha}{2}\right)$ Purely imaginary in nature
- $(z + 16)(\bar{z} + 16) = 16(z + 1)(\bar{z} + 1)$
 $\Rightarrow z\bar{z} = 16 \Rightarrow |z|^2 = 16 \Rightarrow |z| = 4$
- $= \frac{((1+c)^2 - s^2) + i(s(1+c) + s(1+c))}{(1+c)^2 + s^2}$
 $= \frac{2c(c+1) + i2s(c+1)}{2+2c} = c + is$
- $\Rightarrow \frac{2|z|^2 + 2|z| - 3}{|z| + 1} > 3$
 $\Rightarrow (|z| - 2)(2|z| + 3) > 0 \Rightarrow |z| > 2$
- If $|z| = |z - 1|$ Then $|z|^2 = |z - 1|^2 \Rightarrow z + \bar{z} = 1$
 If $|z| = |z + 1|$ Then
 $|z|^2 = |z + 1|^2 \Rightarrow z + \bar{z} = -1 \Rightarrow z + \bar{z} = 1$
- $|3| = |a_0z^n + a_1z^{n-1} + a_2z^{n-2} + \dots + a_{n-1}z + a_n|$
 $3 \leq |a_0||z|^n + |a_1||z|^{n-1} + \dots + a_{n-1}|z| + |a_n|$
 $\Rightarrow 3 < 2(|z|^n + |z|^{n-1} + \dots + |z| + 1)$ if
 $|z| < 1 \Rightarrow \frac{1}{1-|z|} > \frac{3}{2} \Rightarrow 3|z| - 1 > 0 \Rightarrow |z| > \frac{1}{3}$
- $(a - ib)^5 = \alpha - i\beta \quad (-i^2 a - ib)^5 = \alpha - i\beta$
 $\Rightarrow (b + ai)^5 = -\frac{\alpha}{i} + \beta = \alpha i + \beta$
- Use $|z_1z_2| = |z_1||z_2|$
- $z = a + \bar{a}$ = always real $\Rightarrow \arg z = 0$ or π .

$$18. z_1 = \cos \alpha - i \sin \alpha; \quad z_2 = \cos \beta + i \sin \beta$$

$$\Rightarrow |z_1 + z_2| + |z_1 - z_2| = 2 \cos \left(\frac{\alpha - \beta}{2} \right) + 2 \sin \frac{\alpha - \beta}{2} = 2 [\cos \theta + \sin \theta] \leq 2\sqrt{2}$$

$$19. \Rightarrow (z+16)(\bar{z}+16) = 16(z+1)(\bar{z}+1)$$

$$\Rightarrow z\bar{z} = 16 \Rightarrow |z|^2 = 16 \Rightarrow |z| = 4$$

$$20. \text{ For any } z_1, z_2 \in \mathbb{C}, |az_1 - bz_2|^2 + |bz_1 + az_2|^2 \\ = (a^2 + b^2)(|z_1|^2 + |z_2|^2)$$

$$21. w = \frac{z-1}{z+1} \Rightarrow z = \frac{1+w}{1-w}; |z|=1 \Rightarrow |1+w|=|1-w| \\ \Rightarrow w \text{ lies on imaginary axis } \therefore \operatorname{Re}(w) = 0$$

$$22. \left(\frac{1+i}{1-i} \right)^n = \frac{2}{\pi} \sin^{-1} \left(\frac{x^2+1}{x} \right), x > 0 \\ \Rightarrow \left(\frac{1+i}{1-i} \right)^n = \frac{2}{\pi} \cdot \frac{\pi}{2} \Rightarrow \text{least value of } n \text{ is } 4$$

More than one correct answer type questions

$$23. \text{ Let } x = |a + \sqrt{a^2 - b^2}| + |a - \sqrt{a^2 - b^2}| \text{ then} \\ x^2 = 2|a|^2 + 2|a^2 - b^2| + 2|a^2 - (a^2 - b^2)| \\ = 2|a|^2 + 2|b|^2 + 2|a^2 - b^2| \\ = (|a+b|^2 + |a-b|^2 + 2|a+b||a-b|) \\ = (|a+b|+|a-b|)^2$$

$$\therefore x = |a+b| + |a-b| \leq 2|a| + 2|b|$$

$$24. \sqrt{20i-21} = \sqrt{(-25)+4+2(5)(2i)} = \sqrt{(5i)^2 + 2^2 + 2 \times 2 \times 5i} = \pm(2+5i) \\ \text{and } \sqrt{21+20i} = \sqrt{25+(2i)^2 2(5)(2i)} \\ = \sqrt{(5+2i)^2} = \pm(5+2i) \\ \Rightarrow z = 7+7i, -3+3i, -7-7i, 3-3i$$

$$25. 2 \arg z^{3/8} = \arg(z^2 + \bar{z}\sqrt{z}) \Rightarrow \frac{z^2 + \bar{z}\sqrt{z}}{z^{3/4}} \text{ is real} \\ \Rightarrow z^{5/4} + \bar{z}z^{-1/4} = z^{-5/4} + \bar{z}z^{-1/4} \\ \Rightarrow \left(z^{5/4} - \bar{z}z^{-1/4} \right) = \frac{z^{5/4} - \bar{z}z^{-1/4}}{(z\bar{z})^{1/4}} \\ \Rightarrow z^{5/4} = \bar{z}z^{-1/4} \text{ (or) } (z\bar{z})^{1/4} \text{ and hence } z = \bar{z} \text{ or } z\bar{z} = 1$$

$$26. (z_1 + z_2) = |x_1 + x_2| + |y_1 + y_2| \leq (z_1) + (z_2)$$

$$27. \sqrt{z_1}(1-z_2) = \sqrt{z_2}i(1+z_1) \Rightarrow \frac{\sqrt{z_1}}{\sqrt{z_2}} =$$

purely imaginary $\Rightarrow \frac{z_1}{z_2} =$ purely real (negative)

$$\frac{\sqrt{z_1}}{\sqrt{z_2}} + \frac{\sqrt{z_1}}{\sqrt{z_2}} = 0$$

$$\Rightarrow \text{Also } \sqrt{z_1} - i\sqrt{z_2} = \sqrt{z_2}\bar{z}_2\sqrt{z_1} + i\sqrt{z_1}\bar{z}_1\sqrt{z_2}$$

$$\Rightarrow (1-i\sqrt{z_1}\bar{z}_2)(\sqrt{z_1}-i\sqrt{z_2}) = 0$$

$$\Rightarrow \bar{z}_1\bar{z}_2 = -1 \text{ or } \frac{z_1}{z_2} = -1$$

$$28. \arg \left(\frac{z-(3+4i)}{z-(1-2i)} \right) = \frac{\pi}{2}$$

$$\text{So } z_1 = \frac{3+4i+1-2i}{2} = 2+i = z_1$$

$$\Rightarrow |z_1|^8 = 625 \Rightarrow \arg(z_1 - 2) = \frac{\pi}{2}$$

$$29. |z_3| = \frac{|z_2||z_1 - z_4|}{|z_1z_4 - 1|} = \frac{|z_1 - z_4|}{|z_1z_4 - 1|} \leq 1$$

$$\text{if and only if } |z_1 - z_4|^2 \leq |z_1z_4 - 1|^2$$

$$\Rightarrow (z_1 - z_4)(\bar{z}_1 - \bar{z}_4) \leq (\bar{z}_1z_4 - 1)(z_1\bar{z}_4 - 1)$$

$$\Rightarrow (|z_1|^2 - 1)(|z_4|^2 - 1) \geq 0 \text{ as } |z_1| < 1 \Rightarrow |z_4| \leq 1$$

$$30. |z_1| < 1 \text{ are } |z_2| < 2$$

$$\text{a) } (|z_1| - 2|z_2|) \leq |z_1 - 2z_2| \leq |z_1| + 2|z_2| \\ \Rightarrow 3 \leq |z_1 - 2z_2| \leq 5$$

$$\text{b) } |z_1| - |z_2| \leq |z_1 + z_2| \leq |z_1| + |z_2| \Rightarrow \leq |z_1 + z_2| \leq 3$$

$$\text{c) } |z_1 - 3z_2| \geq (|z_1| - 3|z_2|) = 5$$

$$\text{d) } |z_1 - z_2| \geq (|z_1| - |z_2| = 1)$$

$$31. \operatorname{Re}(z_1\bar{z}_2) = 0 \Rightarrow ac + bd = 0 \Rightarrow \frac{a}{b} = \frac{-d}{c} = \alpha \text{ from}$$

$$(1) \text{ and } (2), b^2\alpha^2 + b^2 = c^2\alpha^2 + c^2 \Rightarrow b^2 = c^2$$

$$a^2 = d^2; |w_1| = \sqrt{a^2 + c^2} = \sqrt{c^2 + d^2} = 1;$$

$$|w_2| = \sqrt{b^2 + d^2} = \sqrt{c^2 + d^2} = 1$$

$$\operatorname{Re}(w_1\bar{w}_2) = ab + cd = b\alpha b + c(-c\alpha) = \alpha(b^2 - c^2) = 0$$

Linked comprehension type questions

Passage - I

$$32.$$

$$33. \quad z + \bar{z} = \frac{2\lambda}{1-\gamma} \text{ and } 1-\gamma = \frac{1}{1+|z|^2}.$$

$$\text{Hence } \alpha = \frac{z + \bar{z}}{2(1+|z|^2)}$$

$$34. \quad z - \bar{z} = \frac{2i\beta}{1-\gamma} \Rightarrow \beta = \frac{(z - \bar{z})}{2i(1+|z|^2)} = \frac{(\bar{z} - z)}{2(1+|z|^2)}$$

Passage-II:

35. Let $z_1 = \cos \alpha + i \sin \alpha$, $z_2 = \cos \beta + i \sin \beta$ be the two roots of $z^2 + pz + q = 0$; $z_1 z_2 = q$ and $|q| = |z_1 z_2| = 1$

$$36. \quad z_1 + z_2 = -p \Rightarrow |p| = |z_1 + z_2| \leq |z_1| + |z_2| = 2 \\ \Rightarrow |p| \leq 2$$

37. Also $z_1 z_2 = q$;

$$\arg q = \arg(z_1 z_2) = \arg z_1 + \arg z_2 = \alpha + \beta \dots (1)$$

$$-p = z_1 + z_2 = (\cos \alpha + \cos \beta) + i(\sin \alpha + \sin \beta)$$

$$\Rightarrow -p = 2 \cos \left(\frac{\alpha + \beta}{2} \right) \left[\cos \frac{\alpha - \beta}{2} + i \sin \frac{\alpha - \beta}{2} \right]$$

$$\arg(-p) = \frac{\alpha + \beta}{2} \Rightarrow \pi + \arg p = \frac{\alpha + \beta}{2}$$

$$\Rightarrow 2\pi + 2\arg p = \alpha + \beta \therefore \alpha + \beta = 2\arg p \dots (2)$$

Passage-III:

$$38. \quad z = \frac{(a+ib)(a+ic)}{2a+i(b+c)}; \text{ So that}$$

$$x^2 + y^2 = z\bar{z} = \frac{(a^2 + b^2)(a^2 + c^2)}{4a^2 + (b+c)^2}$$

$$39. \quad \Rightarrow (x-a)^2 + y^2 = (x-a+iy)(x-a-iy) \\ = \frac{(a^2 + bc)^2}{4a^2 + (b+c)^2}$$

$$40. \quad (x^2 + y^2) - [(x-a)^2 + y^2] = 2ax - a^2 \\ \Rightarrow x = \frac{a(2a^2 + b^2 + c^2)}{4a^2 + (b+c)^2}$$

Passage-IV:

$$41. \quad \left. \begin{aligned} a &= \cos \alpha + i^0 \sin \alpha & b &= \cos \beta + i^0 \sin \beta & c &= \cos \gamma + i^0 \sin \gamma \\ \frac{1}{a} \cos \alpha - i \sin \alpha &= \cos \beta - i \sin \beta & \frac{1}{b} &= \cos \gamma - i \sin \gamma \end{aligned} \right\}$$

$$a + b + c = 0$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$$

42. z_1 And z_2 have same direction $\arg z_1 = \arg z_2$

43. $\arg z_1 + \arg z_2 = 0$ and $\arg z_3 + \arg z_4 = 0$

Matrix matching type questions

$$44. \quad A) \quad 1 + z^3 = 1 + \cos 3\alpha + i \sin 3\alpha$$

$$= 2 \cos^2 \frac{3\alpha}{2} + 2i \sin \frac{3\alpha}{2} \cos \frac{3\alpha}{2}$$

$$\Rightarrow \arg(1 + z^3) = 3\alpha / 2$$

$$\text{since } \cos \left(\frac{3\alpha}{2} \right) > 0$$

$$B) \quad 1 - z^4 = 1 - \cos 4\alpha - i \sin 4\alpha$$

$$= 2 \sin^2 2\alpha - 2i \cos 2\alpha \sin 2\alpha$$

$$\Rightarrow \arg(1 - z^4) = 2\alpha - \pi / 2 \text{ since } \sin 2\alpha > 0$$

$$C) \quad \frac{1+z^3}{1-z^4} = \frac{\cos(3\alpha/2)}{\sin(2\alpha)} \left[\cos \left(\frac{\pi - \alpha}{2} \right) + i \sin \left(\frac{\pi - \alpha}{2} \right) \right]$$

$$\Rightarrow \arg \left(\frac{1+z^3}{1-z^4} \right) = \frac{\pi - \alpha}{2}$$

$$D) \quad \frac{z^4 - 1}{z^3 + 1} = \frac{\sin(2\alpha)}{\cos(3\alpha/2)} \left[\cos \left(\frac{\pi}{2} + \frac{\alpha}{2} \right) + i \sin \left(\frac{\pi}{2} + \frac{\alpha}{2} \right) \right]$$

$$\text{Thus, } \arg \left(\frac{z^4 - 1}{z^3 + 1} \right) = \frac{\pi}{2} + \frac{\alpha}{2}$$

$$45. \quad A) \quad 4 \arg \left(\frac{z_1}{z_2} \right) = 4 \left(\frac{\pi}{4} - \frac{\pi}{6} \right) = \frac{\pi}{3}$$

$$B) \quad z = \sqrt{2} \cos \alpha + i(1 + \sqrt{2} \sin \alpha)$$

$$C) \quad \bar{z} + i\omega = 0 \Rightarrow z - i\omega = 0 \Rightarrow z = i\omega;$$

$$\arg z + \arg w = \pi \quad \omega = -iz \quad \arg z - \frac{\pi}{2} + \arg z = \pi$$

D) z^2 and 1 have opposite directions $\arg(z^2) = \pi$
etc.....

Integer answer type questions

$$46. \quad \text{We have } \frac{9}{|z|^2} = (2 + \cos \theta)^2 + \sin^2 \theta = 5 + 4 \cos \theta \dots (1)$$

$$\text{and } \frac{3}{z} + \frac{3}{\bar{z}} = 4 + 2 \cos \theta \dots (2)$$

Eliminating θ from (1) and (2), we get

$$\frac{9}{|z|^2} - 6 \left(\frac{1}{z} + \frac{1}{\bar{z}} \right) = -3 \Rightarrow 3 = 2(\bar{z} + z) - |z|^2.$$

$$47. \quad z = \cos \theta + i \sin \theta, 0 < \theta < 2\pi;$$

$$\left| \frac{z}{\bar{z}} + \frac{\bar{z}}{z} \right| = 1 \Rightarrow 2|\cos 2\theta| = 1 \Rightarrow \cos 2\theta = \frac{1}{2} \text{ or } -\frac{1}{2}$$

$$\Rightarrow \theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6} \text{ or } \Rightarrow \theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$$

48. $z^2 = \bar{z}^3 \Rightarrow |z| = 0$ or $|z| = 1$. If $|z| = 0$, then $z = 0$.

If $|z| = 1$, we get $\bar{z} = 1/z$ and the equation becomes

$$z^2 = \frac{1}{z^3} \Rightarrow z^5 = 1$$

49. $z^4 = i|z|^3 \Rightarrow |z| = 0$ or $|z| = 1$

$$|z| = 0 \Rightarrow z = 0 ; |z| = 1 \Rightarrow z^4 = i$$

50. $\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2} + \frac{\gamma^2}{c^2} + 2\left(\frac{\alpha\beta}{ab} + \frac{\beta\gamma}{bc} + \frac{\gamma\alpha}{ca}\right) = 2i$ and

$$\frac{abc}{\alpha\beta\gamma} \left(\frac{\beta\gamma}{bc} + \frac{\alpha\gamma}{ac} + \frac{\alpha\beta}{ab} \right) = 0$$

51. $\Delta = (a+ib)^2 - 4(c+id) = 0$

$$\Rightarrow (a^2 - b^2 - 4c) + i(2ab - 4d) = 0$$

$$\Rightarrow a^2 = b^2 + 4c \text{ and } ab = 2d \Rightarrow \frac{ab}{d} = 2$$

(Since $bd \neq 0 \Rightarrow b \neq 0$ and $d \neq 0$ and hence $a \neq 0$).

EXERCISE-II

LEVEL-I (MAIN)

Single answer type questions

1. $\left[(x+iy)^n + (x-iy)^n \right] = 2 \left(x^2 + y^2 \right)^{n/2} \cos \left(n \tan^{-1} \frac{y}{x} \right)$

2. $\frac{x}{y} - \frac{y}{x} = \text{cis}(\alpha - \beta) - \text{cis}(\beta - \alpha) = 2i \sin(\alpha - \beta)$

3. $(1+i\sqrt{3})^n = 2^n \text{cis} \frac{n\pi}{3} ; (1-i\sqrt{3})^n = 2^n \text{cis} \left(\frac{-n\pi}{3} \right)$

$$(1+i\sqrt{3})^n + (1-i\sqrt{3})^n = 2^n \cdot 2 \cos \frac{n\pi}{3} = 2^{n+1} \cos \frac{n\pi}{3}$$

4. $a+ib = \sqrt{a^2+b^2} \text{cis} \tan^{-1} \left(\frac{b}{a} \right) ;$

$$a-ib = \sqrt{a^2+b^2} \text{cis} \left(-\tan^{-1} \frac{b}{a} \right)$$

$$\Rightarrow (a+ib)^{m/n} + (a-ib)^{m/n}$$

$$= 2(\sqrt{a^2+b^2})^{m/n} \cdot \cos \left(\frac{m}{n} \tan^{-1} \frac{b}{a} \right)$$

5. Product all values of

$$z^{1/n} = (-1)^{n-1} z = (-1)^{4-1} (-\omega^2)^3 = 1$$

6. $2^n \cdot \text{cis} \frac{n\pi}{3} = 2^n \Rightarrow \cos \frac{n\pi}{3} = 1$

$$\text{and } \sin \frac{n\pi}{3} = 0 \Rightarrow n = 6m$$

7. $\text{cis} \alpha_1 \cdot \text{cis} \alpha_2 \cdot \text{cis} \alpha_3 \dots = \text{cis}(\alpha_1 + \alpha_2 + \alpha_3 \dots)$

$$= \text{cis}(\pi) \left(\frac{1}{2} + \frac{1}{2^2} + \frac{1}{2^3} + \dots \right) = \text{cis} \pi = -1$$

8. $1 + \text{cis} \frac{\pi}{15} + \text{cis} \frac{5\pi}{15} + \text{cis} \frac{7\pi}{15} + \dots + \text{cis} \frac{29\pi}{15}$

$$= 1 + 0 = 1 \text{ sum of } 15^{\text{th}} \text{ root of unity}$$

9. $x = \text{cis} \alpha, y = \text{cis} \beta ; z = \text{cis} \gamma \Rightarrow x + y + z = 0$

$$\Rightarrow x^3 + y^3 + z^3 = 3xyz \Rightarrow \frac{x^3}{xyz} + \frac{y^3}{yzx} + \frac{z^3}{zxy} = \frac{(x^3 + y^3 + z^3)}{xyz} = 3$$

$$\text{impart} = 0 \sin(2\alpha - \beta - \gamma) + \sin(2\beta - \gamma - \alpha)$$

$$+ \sin(2\gamma - \beta - \alpha) = 0 \Rightarrow x^0 = 1$$

10. $\cos 3A + \cos 3B + \cos 3C = 3 \cos(A+B+C) = -z$

11. $x^{12} - 1 = 0 \Rightarrow (x^6)^2 - 1 = (x^6 - 1)(x^6 + 1)$

$$= (x^2 - 1)(x^4 + x^2 + 1)(x^6 + 1)$$

12. From 6 $(-1)^{n-1} \cdot 1$

13. $\frac{x^n - 1}{x - 1} = (x - \alpha_1)(x - \alpha_2) \dots (x - \alpha_{n-1})$ as

$$\Rightarrow x \rightarrow 1 \quad (1 - \alpha_1)(1 - \alpha_2) \dots (1 - \alpha_{n-1}) = n$$

14. Take $x = -1$ in (16)

$$s = 1 + 2\alpha + 3\alpha^2 + \dots + n\alpha^{n-1}$$

15. $\frac{\alpha s = \alpha + 2\alpha^2 + \dots + n\alpha^n}{(1-\alpha)s = 1 + \alpha + \alpha^2 + \dots + \alpha^{n-1} - n}$

$$\therefore \alpha^n = 1 ; s = -\frac{n}{1-\alpha}$$

16. $x = \omega \text{ or } \omega^2 ; \omega^3 = 1 ; \omega + \omega^2 = -1$

17. $\frac{a\omega + b\omega^2 + c\omega^3}{c\omega + a\omega^2 + b} + \frac{a + b\omega + c\omega^2}{b + c\omega + a\omega^2}$

$$= \frac{-a\omega^2 - b - c\omega}{b + c\omega + a\omega^2} = -1$$

18. $\sum_{k=1}^n (n^3 - 1) = \frac{n^2(n+1)^2}{4} - 1^3 - [n-1]$

19. ω, ω^2 Are roots of $\frac{1}{a+x} + \frac{1}{b+x} + \frac{1}{c+x} = \frac{2}{x} ;$

take $x = 1$

20. $z^4 + z^3 + z^2 + z + 1 = 0 \Rightarrow (z^2 + 1)(z^2 + z + 1) = 0$
 $\Rightarrow z^2 = -1$ or $z^3 = 1 \Rightarrow |z| = 1$

21. $|z| = 1 \Rightarrow \bar{z} = \frac{1}{z} \Rightarrow z^n + \bar{z}^n = 2 \cos(n \arg z)$

22. $\because \omega$ is cube root of unity

$$\therefore 1 + \omega + \omega^2 = 0 \Rightarrow 1 + \omega = -\omega^2$$

$$\text{Now if } (1 + \omega)^7 = l + m\omega \Rightarrow -\omega^2 = l + m\omega$$

$$\Rightarrow 1 + \omega = l + m\omega \text{ can comparison } l = 1, m = 1$$

23. $w = \frac{-1 + i\sqrt{3}}{2}, w^3 = 1, \text{ and } \Delta = 3w - 3w^2$

24. $\sum_{r=1}^{10} (r-w)(r-w^2) = 450$ since $w + w^2 = -1, w^3 = 1$

25. $w + w^2 = -1, w^3 = 1$

26. $z^5 = (z-1)^5 \Rightarrow |z| = |z-1| = 1 \Rightarrow \operatorname{Re}(z) = \frac{-1}{2}$

$$\therefore \text{The roots lie on line } \operatorname{Re} z = \frac{-1}{2}$$

27. $\log_2 |(\alpha + \alpha^2 + \alpha^3 + 1) - \alpha^4|$
 $\Rightarrow \log_2 |-2\alpha^4| = \log_2 2 = 1$

28. $z^{100} - 1 = (z-1)(z-\alpha_1)(z-\alpha_2) \dots (z-\alpha_{99}) = 0$

$$\text{Sum of roots} = 1 + \alpha_1 + \alpha_2 + \dots + \alpha_{99} = 0$$

$$\text{Sum of roots taken two at a time} = 0$$

$$\Rightarrow \sum_{1 \leq i < j \leq 99} \alpha_i \alpha_j + \alpha_1 + \alpha_2 + \dots + \alpha_{99} = 0 \Rightarrow \sum_{1 \leq i < j \leq 99} \alpha_i \alpha_j = 1$$

29. $\alpha = \operatorname{cis} x, \beta = \operatorname{cis} y, \gamma = \operatorname{cis} z$ Given

$$\alpha + 2\beta + 3\gamma = 0 \Rightarrow \alpha^3 + 8\beta^3 + 27\gamma^3 = 18\alpha\beta\gamma$$

$$\Rightarrow \sin 3x + 8 \sin 3y + 27 \sin 3z = 18 \sin(x+y+z)$$

Numerical value type questions

30. $z + \frac{1}{z} = 2 \cos 6^\circ \Rightarrow z = \cos 6^\circ \pm i \sin 6^\circ$

$$z^{1000} + \frac{1}{z^{1000}} + 1 = 2 \cos 6000^\circ + 1 = 2 \times \left(-\frac{1}{2}\right) + 1 = 0$$

31. $(\sqrt{3} + i)^{100} = 2^{100} e^{i \frac{\pi}{6} \cdot 100} = 2^{100}$

$$\left(\frac{-1}{2} + \frac{i\sqrt{3}}{2}\right) = 2^{99}(-1 + i\sqrt{3})$$

$$\text{here } a = -1, b = \sqrt{3}$$

32. If $n \neq 3m : \omega^n + \omega^m + (\omega^2)^n = 0 ; \sin\left(-\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$

33. $a + b + c = e,$

$$a + b\omega + c\omega^2 = e^\omega, a + b\omega^2 + c\omega = e^{\omega^2}$$

$$a^3 + b^3 + c^3 - 3abc = (a+b+c)$$

$$(a+b\omega + c\omega^2)(a+b\omega^2 + c\omega)$$

$$= e^{1+\omega+\omega^2} = e^0 = 1$$

34. $\operatorname{cis} \frac{\pi}{6} = -i\omega, \left(2 \operatorname{cis} \frac{\pi}{6}\right)^n = (-2i\omega)^n = (8i)^{\frac{n}{3}}$

$$\left(4 \operatorname{cis} \frac{\pi}{4}\right)^n = (2\sqrt{2}(1+i))^n = (8(1+i)^{\frac{1}{2}})^{\frac{n}{2}} = (16i)^{\frac{n}{4}}$$

$$(8i)^{\frac{n}{3}} = (16i)^{\frac{n}{4}}$$

Which satisfy only $m = 48, n = 24$

$$\left(\frac{n}{m}\right)^2 = \frac{1}{4} = 0.25$$

LEVEL-II (ADVANCED)

Single answer type questions

1. $(x-1)^3 + 8 = 0 \Rightarrow \left(\frac{x-1}{-2}\right)^3 = -1 \Rightarrow \left(\frac{x-1}{-2}\right) = 1,$

$$\omega, \omega^2 \Rightarrow x = -1; 1 - 2\omega; 1 - 2\omega^2$$

$$\therefore \text{required expression} = 3\omega^2.$$

2. $x = -p; x^{1/3} = p^{1/3}(-1)^{1/3} \Rightarrow \alpha = -p^{1/3}; \beta = -p^{1/3}\omega;$

$$\gamma = -p^{1/3}\omega^2 = \frac{1}{w} \frac{xw + yw^2 + z}{w \cdot xw + yw^2 + z} = w^2$$

3. $x^3 + 2x^2 + 2x + 1 = 0 \Rightarrow x = -1, \omega, \omega^2$ will not satisfy $x^{2012} + x^{2014} + 1 = 0.$

4. $\alpha = 2\omega \Rightarrow \alpha^{2n} + (2\alpha)^n + 2^{2n} = 2^{2n}$

$$[\omega^{2n} + \omega^n + 1] = 0$$

5. $z^7 = -1 \Rightarrow z^{86} + z^{175} + z^{289} = z^1 + (-1) + (-2) = -1$

6. $(a+ib)^{2009} = a^2 + b^2 \Rightarrow 2009$ non-zero complex no's and one (0, 0) i.e., 2010 No. of roots exist

7. If $f(i) = |1 - ai - b + ci + d| = 1$

$$\Rightarrow (1 - b + d)^2 + (c - a)^2 = 1$$

$$\Rightarrow \text{One solution } b = d \text{ and } a = c$$

equation has four real roots

$$\Rightarrow t^{11}(x) = 0 \text{ has distinct roots } \Rightarrow 3a^2 > 8b.$$

8. $11(z^{10} + 10iz^9 + 10iz) = 11 \Rightarrow z^{10} + \frac{10i}{11}z^9 + \frac{10i}{11}z = 1$

$$\Rightarrow |z|^{10} + |z|^9 + |z| \geq 1 \text{ is true for } |z| \geq 1$$

$$\text{also } 11(z^{10} - 1) + 10iz(z^8 + 1) = 0 \text{ is true for } z^{10} = 1 \text{ and } z^8 + 1 = 0; \text{ but have no common root.}$$

9. $\operatorname{Im}(Z^{2m-1}) = \sin(2m-1)\theta$

$$\begin{aligned} \sum_{m=1}^{15} \sin(2m-1)\theta &= \sin\theta + \sin 3\theta + \sin 5\theta + \dots + \sin 29\theta \\ &= \frac{\sin\left(\frac{15 \cdot 2\theta}{2}\right)}{\sin \frac{2\theta}{2}} \sin\left(\frac{\theta+29\theta}{2}\right) = \frac{\sin 15\theta}{\sin \theta} \cdot \sin 15\theta \\ &= \frac{1}{4 \sin 2\theta} \quad (\text{when } \theta = 2^\circ) \end{aligned}$$

10. $a\alpha^2 + b\alpha + 1 = 0$; $\overline{a\alpha^2} + \overline{b\alpha} + \overline{1} = 0$;

$$|\alpha| = 1 \Rightarrow \alpha^2 + \overline{b\alpha} + \overline{1} = 0$$

$$\text{we have } \left. \begin{matrix} \frac{\alpha^2}{b} & \frac{\alpha}{a} & \frac{1}{b} \\ \frac{b}{b} & \frac{1}{a} & \frac{a}{b} \end{matrix} \right\} \frac{\alpha^2}{ba-b} = \frac{\alpha}{1-|a|^2} = \frac{1}{ab-b}$$

$$\Rightarrow |ab-b| = 1-|a|^2 = 1-\frac{1}{4} = \frac{3}{4}$$

11. $(C+iS)(C-iS) = (\alpha + \alpha^2 + \alpha^4)(\alpha^6 + \alpha^5 + \alpha^3)$

$$\Rightarrow C^2 + S^2 = 1 + \alpha^6 + \alpha^4 + \alpha + 1 + \alpha^5 + \alpha^3 + \alpha^2 + 1$$

$$(a^7 = 1)$$

$$\Rightarrow \left(-\frac{1}{2}\right)^2 + S^2 = 2 + (1 + \alpha + \alpha^2 + \alpha^3 + \alpha^4 + \alpha^5 + \alpha^6)$$

$$\Rightarrow S^2 = 2 + 0 - \frac{1}{4} = \frac{7}{4} \Rightarrow S = \frac{\sqrt{7}}{2}$$

More than one correct answer type questions

12. Put $\omega = \cos \frac{2\pi}{11} + i \sin \frac{2\pi}{11}$, so that for $1 \leq k \leq 10$

$$\sin \frac{2\pi k}{11} - i \cos \frac{2\pi k}{11} = -i \left(\cos \frac{2\pi k}{11} + i \sin \frac{2\pi k}{11} \right)$$

$$= -i\omega^k \quad [\text{De Moivre's Theorem}]$$

$$\text{Thus, } S = -i \sum_{k=1}^{10} \omega^k = \frac{-i\omega(1-\omega^{10})}{1-\omega} = \frac{-i(\omega-\omega^{11})}{1-\omega}$$

$$\therefore S = i \Rightarrow S + \overline{S} = 0, S\overline{S} = 1 \text{ and } \sqrt{S} = \pm \frac{1}{\sqrt{2}}(1+i)$$

13. Since $f(x)$ is divisible by

$$x^2 + x + 1, f(\omega) = 0, f(\omega^2) = 0$$

$$\therefore P(\omega^3) + \omega Q(\omega^3) = 0 \Rightarrow P(1) + \omega Q(1) = 0 \dots (1)$$

$$\text{and } P(\omega^6) + \omega^2 Q(\omega^6) = 0 \Rightarrow P(1) + \omega^2 Q(1) = 0 \dots (2)$$

Solving (1) and (2) we obtain $P(1) = 0$ and $Q(1) = 0$.

$\therefore$ Both $P(x)$ and $Q(x)$ are divisible by $x-1$.

$\Rightarrow P(x^3)$ and $Q(x^3)$ are divisible by x^3-1 and hence by $x-1$.

Since $f(x) = P(x^3) + xQ(x^3)$, we get $f(x)$ is divisible by $x-1$.

14. $z^{p+2} - z^p - z^2 + 1 = 0 \Rightarrow (z^p - 1)(z^2 - 1) = 0$

$z^{p-1} = 0$ or $z^2 - 1 = 0$ but they cannot have common roots as p, q are distinct primes

15. $(1+i)^n = C_0 - C_2 + C_4 - C_6 + \dots + i(C_1 - C_3 + C_5 + \dots)$

$$(1-i)^n = C_0 - C_2 + C_4 - C_6 + \dots + i(C_1 - C_3 + C_5 + \dots)$$

$$\Rightarrow C_0 - C_2 + C_4 - C_6 + \dots = \frac{1}{2}[(1+i)^n + (1-i)^n]$$

$$= 2^{n/2} \cos \frac{n\pi}{4} C_1 - C_3 + C_5 - C_7 + \dots$$

$$= \frac{1}{2}[(1+i)^n - (1-i)^n] = iz^{n/2} \cdot \sin \frac{n\pi}{4}$$

Linked comprehension type questions

Passage-I

16. We have $P_1(x) = x$ and

$$\cos(n+1)\theta + \cos(n-1)\theta = 2\cos(n\theta)\cos\theta$$

$$\Rightarrow P_{n+1}(x) + P_{n-1}(x) = 2xP_n(x)$$

$$\Rightarrow P_n(x) = \frac{1}{2x}[P_{n+1}(x) + P_{n-1}(x)]$$

17. Put $x = \cos\theta$ so that $\sqrt{x^2-1} = i\sin\theta$ Now,

$$\left(x + \sqrt{x^2-1}\right)^n + \left(x - \sqrt{x^2-1}\right)^n$$

$$= (\cos\theta + i\sin\theta)^n + (\cos\theta - i\sin\theta)^n = 2\cos(n\theta)$$

$$[\text{using De Moivre's theorem}] = 2P_n(\cos\theta).$$

$$\text{Thus, } \left(x + \sqrt{x^2-1}\right)^n + \left(x - \sqrt{x^2-1}\right)^n = 2P_n(x)$$

Passage-II

18. $\frac{z^n-1}{z-1} = (z-\alpha_1)(z-\alpha_2)\dots(z-\alpha_{n-1})$

$$\Rightarrow \log(z^n-1) - \log(z-1) = \sum_{r=1}^{n-1} \log(z-\alpha_r)$$

D.w.r.t. z and take $z=2$;

$$\frac{n \cdot 2^{n-1}}{2^{n-1}} - \frac{1}{2-1} = \sum_{r=1}^{n-1} \frac{1}{2-\alpha_r} = \frac{n \cdot 2^{n-1} - 2^n + 1}{2^n - 1}$$

$$\begin{aligned}
 19. \text{ Now, } \sum_{r=0}^{n-1} \frac{\bar{a}(\alpha)^r + a(\bar{\alpha})^r + b}{|a| + |a|} &= \sum_{r=0}^{n-1} \frac{\bar{a}(\alpha)^r + a(\bar{\alpha})^r + b}{2|a|} \\
 &= \frac{1}{2|a|} \left\{ \bar{a} \sum_{r=0}^{n-1} (\alpha)^r + a \sum_{r=0}^{n-1} (\bar{\alpha})^r + b \sum_{r=0}^{n-1} 1 \right\} \\
 &= \frac{1}{2|a|} (0 + 0 + nb) = \frac{nb}{2|a|} \\
 20. \quad \alpha + \beta &= z + z^2 + z^3 + \dots + z^n = -1 + 1 + z + z^2 + \dots + z^{2n} \\
 \alpha + \beta &= -1 + 0; \therefore \text{sum of } (2n-1)^{\text{th}} \text{ roots of unity} = 0 \\
 \beta &= z(z + z^3 + \dots + z^{2n-1}) = \alpha z; \alpha + \beta = -1; \\
 \alpha + \alpha z &= 1 \Rightarrow \alpha = \frac{-1}{1+z}; \beta = \frac{-z}{1+z} \\
 \alpha\beta &= \frac{z}{(1+z)^2} = \frac{\text{cis } \frac{2\pi}{2n+1}}{\left(1 + \cos \frac{2\pi}{2n+1} + i \sin \frac{2\pi}{2n+1}\right)^2} \\
 &= \frac{1}{4 \cos^2 \left(\frac{\pi}{2n+1}\right)}
 \end{aligned}$$

Passage-III

21. For a regular polygon $z_1^2 + z_2^2 + \dots + z_n^2 = n z_0^2$ where $z_1, z_2, \dots, z_n$ are vertices and z_0 is the centre
22. $A_1 A_2 = 2 \sin \frac{\pi}{n}$;
 $A_1 A_2 = 2 \sin \frac{\pi}{n} \dots A_1 A_r = 2 \sin(r-1) \frac{\pi}{n}$
 $|A_1 A_2|^2 + |A_1 A_3|^2 + \dots + |A_1 A_n|^2 = \sum_{r=2}^n 2^2 \sin^2(r-1) \frac{\pi}{n}$
 $= 2 - 2 \cos \frac{2\pi}{n} + 2 - 2 \cos \frac{4\pi}{n} + \dots$
 $= 2(n-1) - 2 \left(\cos \frac{2\pi}{n} + \cos \frac{4\pi}{n} + \dots + \cos(0-1) \frac{2}{n} \right)$
 $= 2(n-1) + 2 = 2n. |A_1 A_2| |A_1 A_3| \dots |A_1 A_n|$
 $= 2^{n-1} \sin \frac{\pi}{n} \sin \frac{2\pi}{n} \dots \sin \frac{(n-1)\pi}{n} = n ; \therefore k = 2$

Matrix matching type questions

23. $|\lambda_1 a_1 \omega + \lambda_2 a_2 \omega^2 + \dots + \lambda_n a_n \omega^n|$
 $\leq |\lambda_1| |a_1| |\omega| + |\lambda_2| |a_2| |\omega|^2 + \dots + |\lambda_n| |a_n| |\omega|^n$
 $(\because |\omega| = 1, \lambda_i \geq 0) = \lambda_1 |a_1| + \lambda_2 |a_2| + \dots + \lambda_n |a_n| \dots$

- (1) $< (\lambda_1 + \lambda_2 + \dots + \lambda_n) ; (\because |a_i| < 1) = 1 < 2$
 (Q) $\therefore |\lambda_1 a_1 \omega + \lambda_2 a_2 \omega^2 + \dots + \lambda_n a_n \omega^n| < 1$ (S)
 Also, as $\lambda_i \geq 0$ and $\lambda_1 + \lambda_2 + \lambda_3 + \dots + \lambda_n = 1$ none of λ_i can exceed 1, i
 hus $0 \leq \lambda_i \leq 1$, there from Eq.(1) we get
 $\therefore |\lambda_1 a_1 \omega + \lambda_2 a_2 \omega^2 + \dots + \lambda_n a_n \omega^n| < |a_1|$
 $+ |a_2| + \dots + |a_n|$ (T)
 (D) $z^n - 1 = (z-1)(z-\omega)(z-\omega^2) \dots (z-\omega^{n-1})$
 Putting $z = 2$
 $\Rightarrow 2^n - 1 = (2-1)(2-\omega)(2-\omega^2) \dots (2-\omega^{n-1})$

24. The given equation is $\frac{z^5 + 1}{z - 1} = 0$ which means that z_1, z_2, z_3, z_4 are four out of five roots of unity except 1.
 A) $z_1^4 + z_2^4 + z_3^4 + z_4^4 + 1^4 = 0 \Rightarrow \sum_{j=1}^4 z_j^4 = -1$
 B) $z_1^5 + z_2^5 + z_3^5 + z_4^5 + 1^5 = 5 \Rightarrow \sum_{j=1}^4 z_j^5 = 4$
 C) $z^4 + z^3 + z^2 + z + 1 = (z - z_1)(z - z_2)(z - z_3)(z - z_4)$
 Putting $z = -2$ both sides and we get $\prod_{j=1}^4 (z_j + 2) = 11$
 D) $|z_1 - z_2| = \sqrt{2 + 2 \cos 144^\circ}$ for minimum
 $= 2 \cos 72^\circ = \frac{\sqrt{5}-1}{2}$ whose greatest integer is 0

25. A) $(z+1)^7 + z^7 = 0 ;$
 $z^7 + z^7 + 7C_1 z^6 + 7C_2 z^5 + \dots + 1 = 0 ; \sum_{k=0}^6 \text{Re}(z_k) = -\frac{7}{2}$
 B) $\frac{\alpha}{a} + \frac{\beta}{b} + \frac{\gamma}{c} = 1 + i$, Squaring both the sides
 $\frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2} + \frac{\gamma^2}{c^2} + 2 \left(\frac{\alpha\beta}{ab} + \frac{\beta\gamma}{bc} + \frac{\gamma\alpha}{ca} \right) = 2i$
 $\therefore \frac{\alpha^2}{a^2} + \frac{\beta^2}{b^2} + \frac{\gamma^2}{c^2} = 2i \because \frac{a}{\alpha} + \frac{b}{\beta} + \frac{c}{\gamma} = 0$
 C) $z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = \prod_{j=1}^6 (z - z_j)$
 Put $z = -1 \therefore 7 = \prod_{j=1}^6 (-1 - z_j) = \prod_{j=1}^6 (1 + z_j)$
 D) $|z| = |z-1| \Rightarrow |z|^2 = |z-1|^2$

$$\Rightarrow z\bar{z} = (z-1)(\bar{z}-1) \Rightarrow z + \bar{z} = 1$$

$$\text{Again if, } |z| = |z+1| \Rightarrow |z|^2 = |z+1|^2$$

$$\Rightarrow z + \bar{z} = -1 \Rightarrow |z + \bar{z}| = 1$$

Integer answer type questions

26. $z = \cos \alpha + i \sin \alpha$ is a root, $\bar{z} = \cos \alpha - i \sin \alpha$ is also a root. Thus, $\cos \alpha - i \sin \alpha = \frac{1}{\cos \alpha + i \sin \alpha} = \frac{1}{z}$ is a root.

$$\text{Therefore, } \frac{1}{z^n} + \frac{a_1}{z^{n-1}} + \frac{a_2}{z^{n-2}} + \dots + a_n = 0$$

$$\Rightarrow 1 + a_1 z + a_2 z^2 + \dots + a_n z^n = 0$$

$$\Rightarrow 1 + a_1 (\cos \alpha + i \sin \alpha) + a_2 (\cos 2\alpha + i \sin 2\alpha)$$

$$+ \dots + a_n (\cos n\alpha + i \sin n\alpha) = 0$$

[using DeMoivre's Theorem]

Equating real parts, we get

$$1 + a_1 \cos \alpha + a_2 \cos 2\alpha + \dots + a_n \cos n\alpha = 0$$

$$\Rightarrow |a_1 \cos \alpha + a_2 \cos 2\alpha + \dots + a_n \cos n\alpha| = 1.$$

27. We have $z = \frac{1}{2}(\sqrt{3} - i) = -\frac{1}{2}i(1 + i\sqrt{3}) = i\omega^2$ where $\omega \neq 1$ is a cube root of unity.

$$\therefore z^{89} = (i\omega^2)^{89} = i^{89}\omega^{178} = i\omega \text{ Also, } i^{97} = i^{96}i = i$$

$$\text{Thus, } (z^{89} + i^{97})^{94} = (i\omega + i)^{94} = [i(-\omega^2)]^{94} = -\omega^2$$

$$\text{Also, } z^n = i^n \omega^{2n}$$

$\therefore$ The given equation becomes

$$-\omega^2 = i^n \omega^{2n} \Rightarrow i^n \omega^{2n-2} = -1$$

This is possible if n is of the type $4k+2$ and $2n-2$ is a multiple of 3.

That is, $2(4k+2)-2 = 8k+2$ is multiple of 3. The least value of k for which this is possible is 2.

$$\text{Therefore, } n = 10 \text{ and } \lambda = 10 \Rightarrow \frac{\lambda}{2} = 5.$$

28. We have $x^4 - 1 = (x-1)(x-x_1)(x-x_2)(x-x_3).$

$$\begin{aligned} \text{Thus, } \frac{(\omega^2 - x_1)(\omega^2 - x_2)(\omega^2 - x_3)}{(\omega - x_1)(\omega - x_2)(\omega - x_3)} \\ = \frac{1}{\omega + 1} \frac{(\omega^2 - 1)(\omega^2 - x_1)(\omega^2 - x_2)(\omega^2 - x_3)}{(\omega - 1)(\omega - x_1)(\omega - x_2)(\omega - x_3)} \end{aligned}$$

$$= \frac{1}{\omega + 1} \left[\frac{(\omega^2)^4 - 1}{\omega^4 - 1} \right] = 1$$

$$\begin{aligned} 29. \quad x^5 - x^3 + x^2 - 1 &= 0 \Rightarrow (x^3 + 1)(x^2 - 1) = 0 \\ &\Rightarrow (x+1)^2(x-1)(x^2 - x + 1) = 0 \end{aligned}$$

Roots are $-1, -1, 1, -\omega, -\omega^2$;

$$x^4 - 1 = 0 \Rightarrow (x^2 + 1)(x^2 - 1) = 0$$

Roots are $1, -1, i, -i \therefore$ Common roots are 1 and -1

$$\begin{aligned} 30. \quad z_3 \bar{z}_4 &= (1 + i\sqrt{3})^5 (1 - i\sqrt{3})^4 = (1 + i\sqrt{3})(1 + 3)^4 \\ &\Rightarrow \sqrt{3} \operatorname{Im}(z_3 \bar{z}_4) = 3(2^8) = 768 = \lambda \Rightarrow \left[\frac{\lambda}{100} \right] = 7. \end{aligned}$$

$$31. \quad \frac{y_1}{x_4} = \tan \theta = \sqrt{2} - 1 \Rightarrow \theta = 22.5^\circ$$

$$\text{separation of } z \text{ and } \bar{z} = 2\theta = \frac{2\pi}{n} = \frac{\pi}{4} \Rightarrow n = 8$$

EXERCISE-III

LEVEL-I (MAIN)

Single answer type questions

1. Centroid of ΔABC is at $\frac{z_1 + z_2 + z_3}{3}$. Orthocentre divides line of Centroid and circumcentre in 2:3 externally.

2. The given equation is $\left| \frac{z-i}{z+1} \right| = \frac{m}{2}$, which does not represent a circle when $\frac{m}{2} = 1$.

$$3. \quad x^2 + y^2 = 2; 3 + 4z = 3 + 4x + 4iy$$

$$\Rightarrow \left(\frac{x+3}{4} \right)^2 + \left(\frac{y}{4} \right)^2 = 2 \Rightarrow (x-3)^2 + y^2 = 32$$

$$4. \quad W_i = \frac{z+i}{z-i} \Rightarrow |z+i| = |z-i|$$

z lies on perpendicular bisector of $(0, -1)$ and $(0, 1)$ i.e. x axis (Real axis)

$$5. \quad |2x| + |2iy| = 2 \Rightarrow |x| + |y| = 1 \Rightarrow \text{square}$$

$$6. \quad |z|^2 + 3z = 2 \Rightarrow x^2 + y^2 + 3x = 3 \text{ and } y = 0$$

$$7. \quad \text{base} = \frac{1}{3} < 1 \Rightarrow |z+1| < |z-1| \Rightarrow \operatorname{Re}(z) < 0$$

8. $(6, 4)(8, 4)$ subtends 45° at (x, y)

$$\Rightarrow C(7, 5) \quad r = \sqrt{2} \Rightarrow (x-7)^2 + (y-5)^2 = 2$$

9.  $\Delta = \frac{1}{2}|z| \cdot |z| \sin \frac{2\pi}{3} = \frac{|z|^2 \sqrt{3}}{4} = \frac{9\sqrt{3}}{4}$

10. $z_1^2 + z_2^2 = z_1 z_2 \Rightarrow z_1, z_2$ 0 from equilateral Δ^{ic}

11. $z_1^2 + z_2^2 = z_1 z_2 \Rightarrow (a+i)^2 + (1+bi)^2 = (a+i)(1+bi)$
 $\Rightarrow a^2 - 1 + 2ai + 1 - b^2 + 2bi = a - b + (1+ab)i$
 $\Rightarrow a^2 - b^2 = a - b$ and $2a + 2b = 1 + ab \Rightarrow a = b$ (or)
 $a + b = 1$ and $2(a+b) = 1 + ab$

$$a = b \Rightarrow 4a = 1 + a^2 \Rightarrow a = 2 - \sqrt{3} \quad a + b = 1$$

 $\Rightarrow ab = 1$ is not possible

12. $z + \alpha\beta = \alpha, \alpha\omega, \alpha\omega^2 \Rightarrow z_1, z_2, z_3$ are equidistant from $\alpha\beta$ and circumradius $= |\alpha| \Rightarrow \sin = 2|\alpha| \sin \theta$

13. $|PA - PB| = k$ represents a hyperbola if $K < AB = 2$ and +ve

14. $x = 1 - t; y = \sqrt{t^2 + t + 2} \Rightarrow y^2 = (x-1)^2 + 1 - x + 2$
 $\Rightarrow x^2 - y^2 - 3x + 4 = 0$

Numerical value type questions

15. $2z^2 + 2z + k = 0; z = \frac{-2 \pm \sqrt{4-8k}}{4}$; Since 'z' is a complex number $4 - 8k$ will be negative $\Rightarrow k > 1/2$
 $(0, 0), \left(\frac{-1}{2}, \frac{\sqrt{2k-1}}{2}\right), \left(\frac{-1}{2}, -\frac{\sqrt{2k-1}}{2}\right)$ Since it is equilateral $\therefore \frac{1}{4}(2k-1) + \frac{1}{4} = (2k-1) \Rightarrow k = 2/3$.

16. $\sqrt{2} = 1.41; x = 1 - t; y = \sqrt{t^2 + t + 2}$
 $\Rightarrow y^2 = (x-1)^2 + 1 - x + 2$
 $\Rightarrow x^2 - y^2 - 3x + 4 = 0$

17. $a\alpha^2 + b\alpha + 1 = 0 \quad (1)$
 $\bar{a}\alpha^2 + \bar{b}\alpha + 1 = 0 \quad (2)$
 $\alpha^2 + \bar{b}\alpha + \bar{a} = 0 \quad (\text{from (1) and (2)})$

$$\frac{\alpha^2}{ab - b} = \frac{\alpha}{1 - |a|^2} = \frac{1}{ab - b}$$

 $\Rightarrow |ab - b| = 1 - |a|^2 = 3/4 = 0.75$

18. $\text{Re}(z) = |z| - 2 \Rightarrow y^2 = 4(x-1)$

$$\therefore z = 1 + t^2 + i2t \quad (\text{parametric form})$$

$$\text{Let } \omega = p + iq \Rightarrow \omega = 1 + s^2 + i2s$$

$$z - \omega = t^2 - s^2 + 2i(t-s) \Rightarrow \text{Arg}(z - \omega) = \frac{\pi}{3}$$

$$\tan^{-1}\left(\frac{2}{t+s}\right) = \frac{\pi}{3}$$

$$t+s = \frac{2}{\sqrt{3}} \Rightarrow z + \omega = 2 + t^2 + s^2 + 2i(t+s)$$

$$\text{Im}(z + \omega) = \frac{4}{\sqrt{3}}$$

$$A. \quad \frac{4}{3} = 1.33$$

LEVEL-II (ADVANCED)

Single answer type questions

1. We know that, $\frac{z_2 - z_1}{z_4 - z_1} = \frac{z_2 - z_1}{z_4 - z_1} e^{i\frac{\pi}{2}} = i$
 (as $|z_2 - z_1| = |z_4 - z_1| \Rightarrow (z_4 - z_1)^2 + (z_2 - z_1)^2 = 0$

Similarly $\Rightarrow (z_4 - z_3)^2 + (z_2 - z_3)^2 = 0$ add the results

2. we know that $\frac{z_3 - z_2}{z_4 - z_2} = \frac{z_3 - z_2}{z_4 - z_2} e^{-i\frac{\pi}{2}}$
 $\Rightarrow z_3 - z_2 = -i(z_4 - z_2) \Rightarrow (z_3 - z_2)^2 + (z_4 - z_2)^2 = 0$

3. Clearly the two eqns represent two rays which are not intersecting. Hence no point of intersection.

4. $\frac{2}{z_1} = \frac{1}{z_2} + \frac{1}{z_3} \Rightarrow \arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) = \arg\left(-\frac{z_2}{z_3}\right) = \arg\left(\frac{z_2}{z_3}\right) \pm \pi$
 $\Rightarrow \arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right) - \arg\left(\frac{z_2 - 0}{z_3 - 0}\right) = \pm \pi$

Sum of angles at A and origin is $\pm \pi$. Hence points O, B, A, C are concyclic.

5. Let $z = x + iy = r(\cos \theta + i \sin \theta)$, then

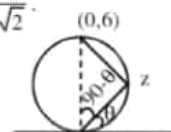
$$|2(x-2) + 2i(y-1)| = r \left(\frac{1}{\sqrt{2}} \cos \theta - \frac{1}{\sqrt{2}} \sin \theta \right)$$

 $= \frac{1}{\sqrt{2}} (r \cos \theta - r \sin \theta) \Rightarrow \sqrt{(x-2)^2 + (y-1)^2} = \frac{1}{\sqrt{2}} \left(\frac{x-y}{\sqrt{2}} \right)$

It is an ellipse with focus at (2,1) and directrix

$$x - y = 0 \text{ and eccentricity } = \frac{1}{\sqrt{2}}$$

6. $\frac{6i - z}{o - z} = \tan(90^\circ - \theta) e^{-i\frac{\pi}{2}}$
 $\frac{-6i}{z} + 1 = (\cot \theta)(-i)$
 $\Rightarrow \cot \theta = \frac{6}{z + i}$



7. The new complex number is

$$2(3+4i)e^{\pi/4} = \sqrt{2}(-1+7i).$$

8. $3z_1 = 5z_2 - 2z_3 \Rightarrow z_1 = \frac{5z_2 - 2z_3}{3}$

z_1 divides line joining z_2 and z_3 externally in 5 : 2

$\Rightarrow z_1, z_2, z_3$ are collinear.

- 9.
- $|z-2i|=1$
- is a circle with centre (0, 2) and radius = 1
-
- complex number of least modulus = (0, 1) =
- $i = e^{i\pi/2}$

$$i^i = e^{-\pi/2} \Rightarrow \sin[\log e e^{-\pi/2}] = \sin\left(\frac{-\pi}{2}\right) = -1$$

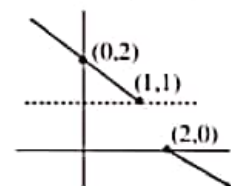
$$\begin{aligned} 10. \quad 1 &= |a|^2 = |b \cos \alpha + c \sin \alpha|^2 \\ &= |b|^2 \cos^2 \alpha + |c|^2 \sin^2 \alpha + (b\bar{c} + \bar{b}c) \cos \alpha \sin \alpha \\ &= 1 + \left(\frac{b}{c} + \frac{c}{b}\right) \cos \alpha \sin \alpha \Rightarrow 1 = 1 + \frac{b^2 + c^2}{2bc} (\sin 2\alpha) \\ &\Rightarrow \frac{b^2 + c^2}{2bc} (\sin 2\alpha) = 0 \Rightarrow b^2 + c^2 = 0 \Rightarrow b = \pm ic \end{aligned}$$

$$\begin{aligned} 11. \quad \bar{z}_1 + \bar{z}_2 + \bar{z}_3 &= 0. \text{ The triangle is equilateral with } R \\ &= 1 \text{ with side} = \sqrt{3} \Rightarrow z_1 z_2 + z_2 z_3 + z_3 z_1 = 0 \\ &\Rightarrow z_1^2 + 2z_2^2 + z = (z_1 + z_3)^2 + 2z_2^2 - 2z_1 z_3 \\ &= 3z_2^2 - 2z_2^2 = z_2^2 \end{aligned}$$

12. put $z = x + iy$

$$\begin{aligned} 13. \quad \text{Put } z &= x + iy \text{ then } \operatorname{Im}\left(\frac{z^2}{z-1}\right) = 0 \\ &\Rightarrow y(x^2 + y^2 - 2x) = 0 \end{aligned}$$

$$14. \quad \operatorname{Arg}(z - (1+i)) = \begin{cases} \frac{3\pi}{4}; & x \leq 1 \\ -\frac{\pi}{4}; & x > 2 \end{cases} \therefore \text{The locus of } z \text{ is}$$



$$\begin{aligned} 15. \quad &\Rightarrow \frac{|z-3|^2 + 2}{11|z-3|-2} < \frac{1}{3} \Rightarrow 3t^2 - 11t + 8 < 0 \\ &\Rightarrow (3t-8)(t-1) < 0 \Rightarrow 1 < t < 8/3 \\ &\Rightarrow 1 < |z-3| < 8/3 \end{aligned}$$

- 16.
- z
- lies on circle with
- $(-1, 1), (5, 1)$
- as ends of diameter.

17. For a regular polygon $z_1^2 + z_2^2 + \dots + z_n^2 = n \cdot z_0^2$

18. Locus is major arc of circle through (0, -1) (0, 1) which subtend
- 90°
- at (1, 0)

$$\Rightarrow \text{radius} = \sqrt{2}, \text{ arc length} = 2\pi\sqrt{2} - \sqrt{2} \frac{\pi}{2} = \frac{3\pi}{\sqrt{2}}$$

19. Image of
- z
- in real axis =
- $\bar{z} \Rightarrow$
- Image of
- $\arg$

$$\left(\frac{z-i}{z-3}\right) = \frac{\pi}{6} \text{ in real axis is } \arg\left(\frac{\bar{z}-i}{\bar{z}-3}\right) = \frac{\pi}{6}$$

$$= \arg\left(\frac{\overline{z+i}}{\overline{z-3}}\right) = \frac{\pi}{6} \Rightarrow -\arg\left(\frac{z+i}{z-3}\right) = \frac{\pi}{6}$$

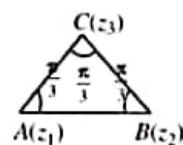
20. $\arg(i\omega) + \arg(i\omega^2) = \arg(i^2 \omega^3) = \arg(-1) = \pi$

21. The locus of z is part of $|z-7-9i| = 3\sqrt{2}$

More than one correct answer type questions

22. I) Suppose
- $\triangle ABC$
- is equilateral (see figure).

$$\frac{z_3 - z_1}{z_2 - z_1} = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$$



Therefore, I $\Rightarrow$ II. This implies (a) is true.

$$\text{II) Assume that } z_3 - z_1 = (z_2 - z_1) \left(\cos \frac{\pi}{3} + i \sin \frac{\pi}{3} \right)$$

$$\text{Therefore } \frac{z_3 - z_1}{z_2 - z_1} = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$$

$$\Rightarrow |z_3 - z_1| = |z_2 - z_1| \text{ and } \angle BAC = \frac{\pi}{3}$$

This implies $\triangle ABC$ is equilateral.

Therefore, II $\Rightarrow$ I. Now rotate $\overline{AC}$ about A through angle

$5\pi/3$ in anticlock sense so that

$$z_2 - z_1 = (z_3 - z_1) \left(\cos \frac{5\pi}{3} + i \sin \frac{5\pi}{3} \right).$$

This means II $\Rightarrow$ III.

Similarly we can see that III $\Rightarrow$ IV and IV $\Rightarrow$ I.

23. As DPQR is equilateral orthocentre, circumcentre centroid will coincide

$$\left| \frac{Z_1 + Z_2 + Z_3}{3} \right| = 1$$

$$\Rightarrow |Z_1 + Z_2 + Z_3|^2 = 9$$

$$\Rightarrow (Z_1 + Z_2 + Z_3)(\bar{Z}_1 + \bar{Z}_2 + \bar{Z}_3) = 9$$

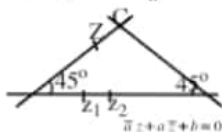
$$\Rightarrow \left(\frac{4}{Z_1} + \frac{1}{Z_2} + \frac{1}{Z_3} \right) \left(\frac{4}{\bar{Z}_1} + \frac{1}{\bar{Z}_2} + \frac{1}{\bar{Z}_3} \right) = 9$$

$$(\because |Z_1|^2 = 4, |Z_2|^2 = 1, |Z_3|^2 = 1) \text{ and } \angle QOR = 120^\circ$$



$$24. \frac{z-c}{|z-c|} = \frac{z_1-z_2}{|z_1-z_2|} e^{\pm i\frac{\pi}{4}}; \frac{z-c}{\bar{z}-\bar{c}} = \frac{z_1-z_2}{\bar{z}_1-\bar{z}_2} e^{\pm i\frac{\pi}{2}}$$

$$\frac{z-c}{\bar{z}-\bar{c}} = \frac{-a}{\bar{a}} (\pm i) \frac{z-c}{a} \pm i \frac{\bar{z}-\bar{c}}{\bar{a}} = 0$$



$$25. \left| z - \frac{z_1 + 2z_2}{3} \right| = \frac{1}{3} |z_1 - z_2| \text{ and}$$

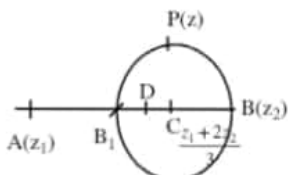
$$\text{Arg} \left(\frac{z_1 - z_2}{z - (kz_1 + (1-k)z_2)} \right) = \pm \frac{\pi}{2}$$

Angle between the line segment joining z_1 and z_2 ;

$$\text{and } kz_1 + (1-k)z_2 \text{ is } \frac{\pi}{2} \quad \frac{z_1 + 2z_2}{3}$$

is a point on segment AB such that AC : CS = 2:1 and

$$D (kz_1 + (1-k)z_2)$$



is a point on segment AB such that AD:DB = 1 - k :

$$k \Rightarrow BD = k |z_1 - z_2|$$

(a) for nosol $BD > BB_1$

$$\Rightarrow k |z_1 - z_2| > \frac{2}{3} |z_1 - z_2| \Rightarrow k > \frac{2}{3}$$

(b) for more than one sol. $0 < BD < BB_1$

$$\Rightarrow 0 < k |z_1 - z_2| < \frac{2}{3} |z_1 - z_2| \quad 0 < k < 2/3.$$

26. Let $|\alpha| = r > 1$ and $\alpha = r \text{cis} \theta$ then

$$z = x + iy = \alpha + \frac{1}{\alpha} = r \text{cis} \theta + \frac{\text{cis}(-\theta)}{r}$$

$$\Rightarrow x = \left(r + \frac{1}{r} \right) \cos \theta \text{ and } y = \left(r - \frac{1}{r} \right) \sin \theta.$$

$$\text{Eliminating } \theta \text{ gives } \frac{x^2}{\left(r + \frac{1}{r} \right)^2} + \frac{y^2}{\left(r - \frac{1}{r} \right)^2} = 1$$

Which is an ellipse. $a = r + \frac{1}{r}$, $b = r - \frac{1}{r}$,

$$(r = |\alpha| > 1 \Rightarrow a > b); \therefore e \sqrt{1 - \frac{b^2}{a^2}} = \frac{2}{r + \frac{1}{r}},$$

distance between foci = $2ae$

$$= 4 \Rightarrow \text{distance between directrices} = \frac{2a}{e}$$

$$27. |z_1 + z_2| = \left| -\frac{b}{a} \right|, z_1 z_2 = \frac{c}{a} \therefore |z_1 + z_2|^2 = 1$$

$$\Rightarrow (z_1 + z_2)(\bar{z}_1 + \bar{z}_2) = 1$$

$$\Rightarrow \frac{(z_1 + z_2)^2}{z_1 z_2} = 1 \Rightarrow \frac{b^2}{a^2} = \frac{c}{a}$$

$$\Rightarrow b^2 = ac \quad z_2 = z_1 e^{i\theta} \Rightarrow |z_2 + z_1| = |z_1| |1 + e^{i\theta}|$$

$$\Rightarrow 2 \cos \left(\frac{\theta}{2} \right) = 1 \Rightarrow \theta = \frac{2\pi}{3} \Rightarrow |z_2 + z_1| = \sqrt{3}$$

Linked comprehension type questions

Passage-I

$$\text{avg} \left(\frac{z_3 - z_1}{z_2 - z_1} \right) = \frac{\pi}{2}, \pi - 2A = \frac{\pi}{2} \Rightarrow A = \frac{\pi}{4}.$$



$$EF = a \cos A = 4 \times \cos \frac{\pi}{4} = 2\sqrt{2}$$

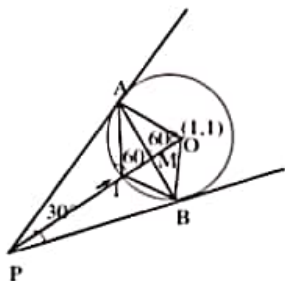
$$28. R = \frac{BC}{2 \sin \frac{\pi}{4}} = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

$$29. EF = a \cos A = 4 \times \cos \frac{\pi}{4} = 2\sqrt{2}$$

$$30. \text{Circum radius of } \triangle DEF, R_1 = \frac{EF}{2 \sin \frac{\pi}{2}} = \frac{2\sqrt{2}}{2} = \sqrt{2}$$

Passage-II

Incentre of ΔPAB lies on same circle



$$\angle AOB = \frac{\pi}{2} + 30^\circ = 120^\circ; OA = 4,$$

$$AM = \frac{4\sqrt{3}}{2} = 2\sqrt{3}, OA = AI = 4$$

31. Area of

$$\Delta IAB = \frac{1}{2} (2\sqrt{4})^2 \sin 120^\circ = \frac{1}{2} \times 4 \times 4 \times \frac{\sqrt{3}}{2} = 4\sqrt{3}$$

32. Circum radius of ΔIAB , $R_1 = \frac{AB}{2 \sin 120^\circ} = \frac{4\sqrt{3}}{2 \cdot \frac{\sqrt{3}}{2}} = 4$

33. Area of $\Delta PAB = PM \times AM = 6 \times 2\sqrt{3} = 12\sqrt{3}$

Passage-III

34. z_1, z_2 lie on the line

$$\Rightarrow \begin{cases} a\bar{z}_1 + \bar{a}z_1 + b = 0 \\ a\bar{z}_2 + \bar{a}z_2 + b = 0 \end{cases} \Rightarrow a(\bar{z}_1 - \bar{z}_2) + \bar{a}(z_1 - z_2) = 0$$

$$\Rightarrow \frac{z_1 - z_2}{\bar{z}_1 - \bar{z}_2} = \frac{-a}{\bar{a}}$$

35. Let z a point on the line

$$\frac{z - z_1}{\bar{z} - \bar{z}_1} = \mu \Rightarrow z - z_1 = \mu(\bar{z} - \bar{z}_1)$$

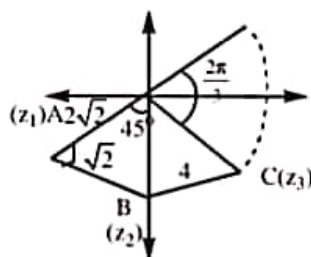
36. Let $z = e^{i\theta}$ be a point on the line with complex

$$\text{slope } e^{i\theta} \quad \frac{e^{i\theta} - 0}{e^{-i\theta} - 0} = e^{i\theta}$$

$$\Rightarrow 2\theta = \theta \Rightarrow \theta = \frac{\phi}{2}$$

Passage-IV (37,38)

$$z_1 = 2\sqrt{2}e^{\frac{3\pi}{4}} = -2 - 2i \quad \frac{z_2 - (2 - 2i)}{0 - (-2 - 2i)} = \frac{1}{2}e^{\frac{i\pi}{2}}$$



$$\Rightarrow z_2 = -1 - 3i \Rightarrow z_3 = 3 - 3i$$

$$\frac{z_4 - 0}{3 - 3i - 0} = e^{\frac{2\pi}{3}} \Rightarrow z_4 = 3(1 - i)W$$

Matrix matching type questions

39. A) $x^2 - y^2 = 2x \Rightarrow (x - 1)^2 - y^2 = 1$ a rectangular hyperbola

B) $PA + PB = k; k > AB \Rightarrow$ locus of P is ellipse

C) $|z - i/2| = m|z + 1|; M = 1 \rightarrow$ a straight line
 $M \neq 1 \rightarrow$ a circle

D) $x^2 + y^2 = 625$; locus of $(-1 + 75x, -75y)$ is

$$\frac{(x+1)^2}{75^2} + \frac{y^2}{75^2} = 625$$

40. A) Parallelogram $\rightarrow$ Opposite sides equal and parallel

B) Rectangle $\rightarrow$ Opposite sides equal and parallel and adjacent sides at right angles

C) Rhombus $\rightarrow$ Opp. sides parallel; all sides equal

D) Square $\rightarrow$ all sides equal; adjacent sides at 90°

Integer answer type questions

41. Let $A(1, 2)$ and $B(-1, 4)$ be the given points.

The equation of the perpendicular bisector of $\overline{AB}$ is $x - y + 3 = 0$.

Given line is $z(1 + i) + \bar{z}(1 - i) + K = 0$

$$\Rightarrow z(1 + i) + \overline{z(1 + i)} + K = 0$$

$$\Rightarrow 2 \operatorname{Re}(z(1 + i)) + K = 0$$

$$\Rightarrow 2 \operatorname{Re}[(x + iy)(1 + i)] + K = 0$$

$$\Rightarrow 2(x - y) + K = 0 \Rightarrow x - y + \frac{K}{2} = 0$$

$$\therefore \frac{K}{2} = 3 \Rightarrow K = 6.$$

42. Let $z_1 = x_1 + iy_1, z_2 = x_2 + iy_2$ and $z_3 = x_3 + iy_3$ be the vertices.

Consider

$$\begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix} = \begin{vmatrix} x_1 + iy_1 & x_1 - iy_1 & 1 \\ x_2 + iy_2 & x_2 - iy_2 & 1 \\ x_3 + iy_3 & x_3 - iy_3 & 1 \end{vmatrix} \rightarrow C_1 \rightarrow C_1 + C_2$$

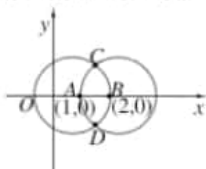
$$= 2 \begin{vmatrix} x_1 & x_1 - iy_1 & 1 \\ x_2 & x_2 - iy_2 & 1 \\ x_3 & x_3 - iy_3 & 1 \end{vmatrix} \rightarrow C_2 \rightarrow C_2 - C_1$$

$$= -2i \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = (-4i) \frac{1}{2} \begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

$$= -4i \times \text{Area of triangle.}$$

$$\therefore \text{Area of triangle} = \frac{i}{4} \begin{vmatrix} z_1 & \bar{z}_1 & 1 \\ z_2 & \bar{z}_2 & 1 \\ z_3 & \bar{z}_3 & 1 \end{vmatrix}; \therefore \lambda = \frac{1}{4} \Rightarrow \frac{1}{\lambda} = 4.$$

43. $|z-1| \leq 1$ represents the interior and boundary of the circle with centre at $1+0i$ and radius 1 and $|z-2|=1$ represents the circle with centre at $2+0i$ and radius 1.



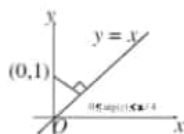
Clearly the points z satisfying $|z-1| \leq 1$ and $|z-2|=1$ lie on the arc DAC.

$$\therefore OA \leq |z| \leq OC (= OD)$$

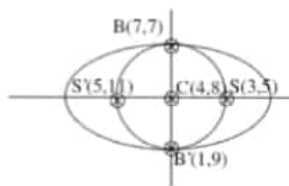
$$\text{As } \angle OCB = \pi/2, OC^2 = OB^2 - BC^2 = 4 - 1 = 3$$

$$\Rightarrow OC = \sqrt{3} \quad \text{Thus, } |z|^2 \leq 3.$$

44. $0 \leq \arg(z) \leq \pi/4$ represents the region of the complex plane lying in the first quadrant and bounded by the x -axis and the line $y=x$. The least value of $|z-i|$ is the length of perpendicular from $(0,1)$ to $y=x$, which is $\frac{1}{\sqrt{2}}$.



Thus, least value of $\sqrt{2}|z-i| = 1$.



45.

1st eq is a circle with centre $(4,8)$ and radius $=\sqrt{10}$.

2nd eq is an ellipse whose foci are $S(3,5), S'(5,11)$.

Centre $C(4,8), a=2\sqrt{5}, ae=\sqrt{10}$ and hence $b=\sqrt{10}$.

$\therefore$ 1st eq is circle on SS' as diameter and hence

$B(7,7)$ and $B'(1,9)$ are the common points.

46. Locus of $\frac{1}{ai-2i} = \frac{a_i+2i}{ai^2+4} \Rightarrow x = \frac{a_i}{ai^2+4}$

$$y = \frac{2}{ai^2+4} \Rightarrow x^2 + y^2 = \frac{1}{ai^2+4} = \frac{y}{2}$$

$$\Rightarrow x^2 + y^2 - \frac{y}{2} = 0 \text{ is a circle of radius } = \frac{1}{2}$$

$$\text{Area} = 8 \times \frac{1}{2} \times \frac{1}{2} \Rightarrow \sin \frac{\pi}{4} = \frac{1}{2} \Rightarrow \sqrt{2}A = 1$$

47. $2W, 2\bar{W}$ are conjugate; $2+3W, 2+3\bar{W}$ are

conjugates $2-(W+W^2)=3$

$$2-(W+W^2)=3 \therefore 5 \text{ roots are minimum}$$

PRACTICE SHEET

EXERCISE-I

LEVEL-I (MAIN)

Single answer type questions

1. $\frac{(3+2i \sin \theta)(1+2 \sin \theta)}{1+4 \sin^2 \theta}$ is purely imaginary

$$\Rightarrow 3-4 \sin^2 \theta = 0 \Rightarrow \theta = n\pi \pm \frac{\pi}{3}$$

2. $(z+1)(\bar{z}+1) = (4+i)(4-i) = 16+1 = 17$

3. $\text{Arg} z_1 + \text{Arg} z_2 = 0$ and $\text{Arg} z_3 + \text{Arg} z_4 = 0$

$$\Rightarrow \text{Arg} z_3 - \text{Arg} z_2 = \text{Arg} z_1 - \text{Arg} z_4$$

$$\Rightarrow \text{Arg} \left(\frac{z_3}{z_2} \right) = \text{Arg} \left(\frac{z_1}{z_4} \right)$$

4. $|z_1 + z_2| = |z_1 - z_2| \Rightarrow \text{Re}(z_1 \bar{z}_2) = 0$

$$\Rightarrow x_1 x_2 + y_1 y_2 = 0 \Rightarrow \text{Arg} \frac{z_1}{z_2} = \frac{\pi}{2}$$